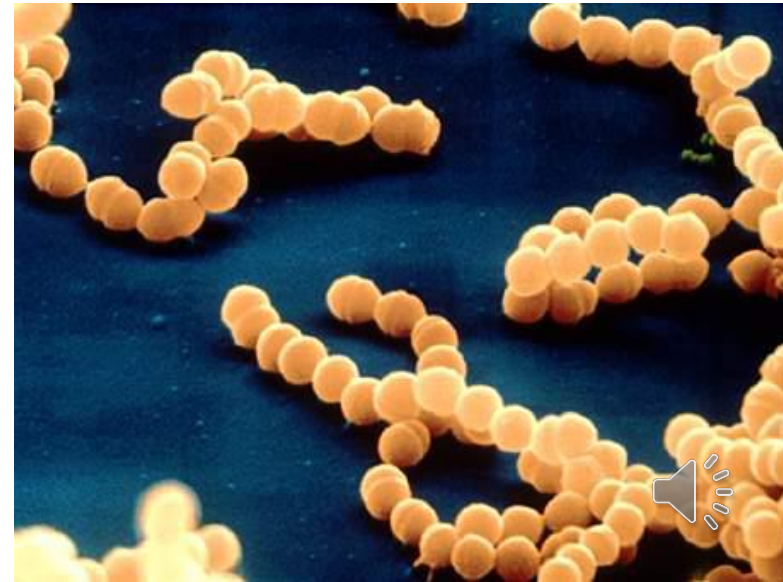
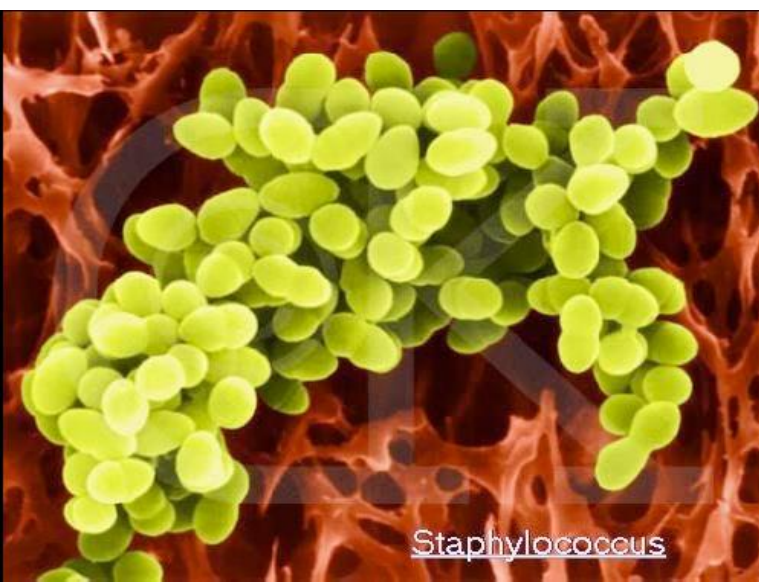


Gram positive cocci

Assoc.Prof.Dr.Theeraphon Piacham



Gram positive vs Gram negative Bacteria

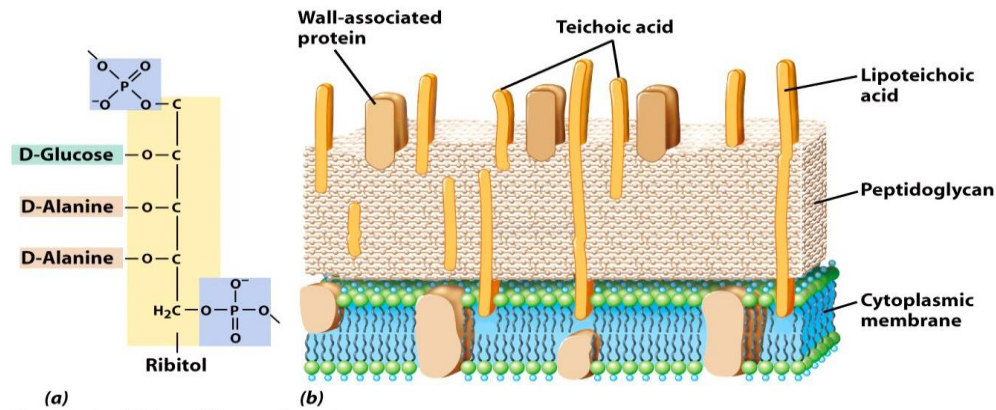


Figure 4-31 Brock Biology of Microorganisms 11/e
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Gram positive bacteria

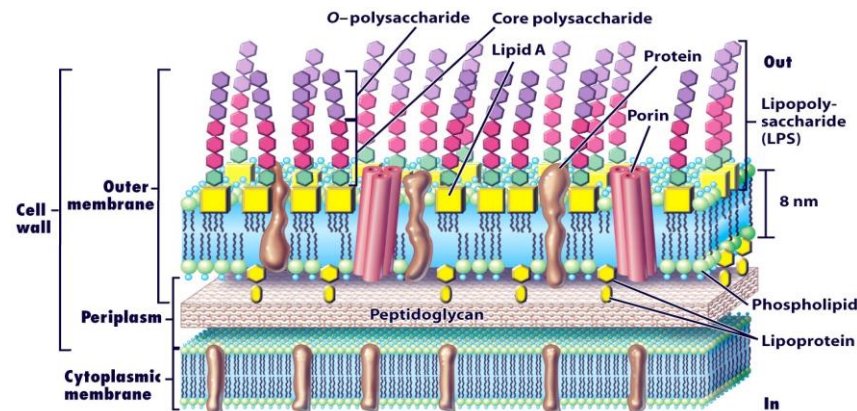
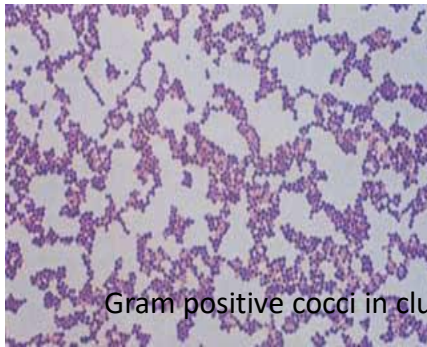


Figure 4-35a Brock Biology of Microorganisms 11/e
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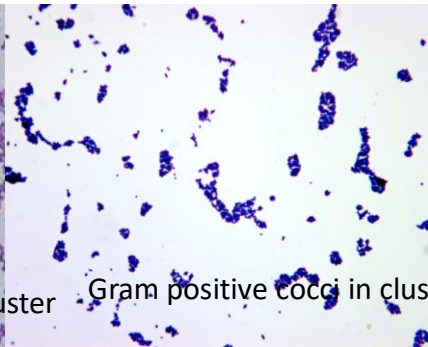
Gram negative bacteria



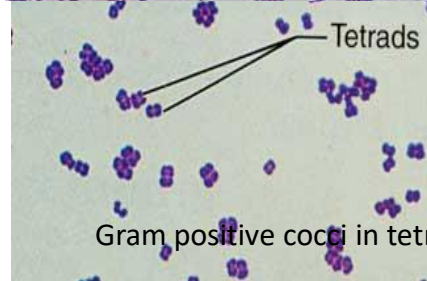
Microscopic examination



Gram positive cocci in cluster



Gram positive cocci in cluster

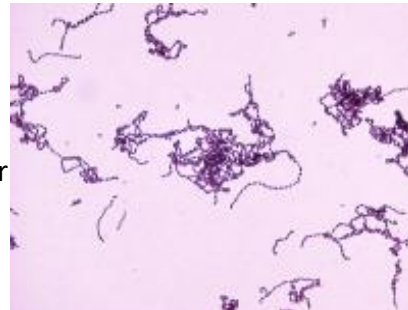


Tetrads

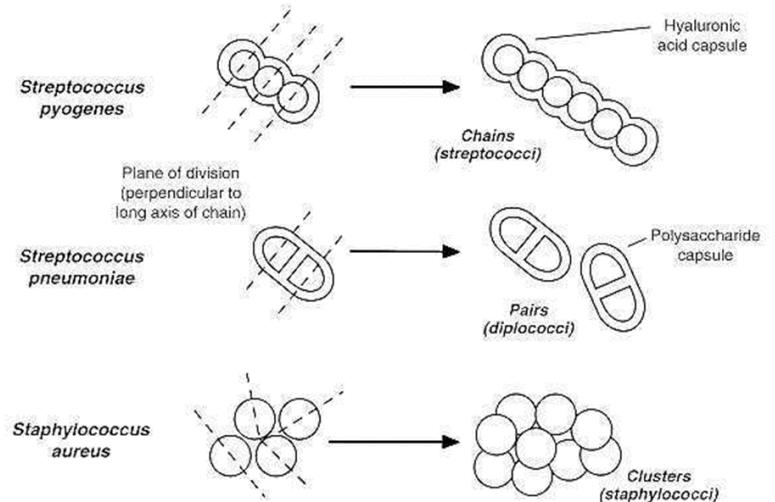
Gram positive cocci in tetrads



Gram positive cocci in chain

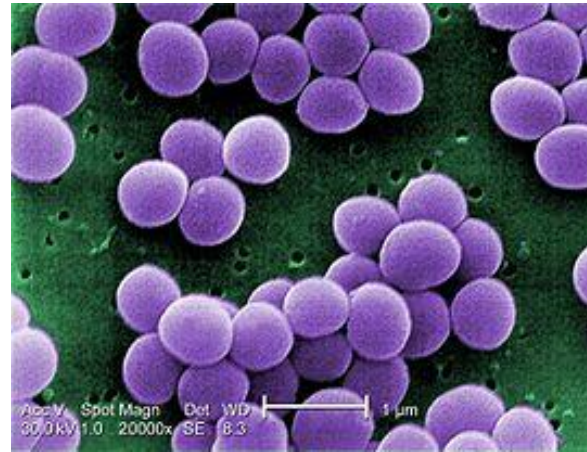


Gram positive cocci in long chain



The gram positive cocci of medical importance

- *Staphylococcus*
- *Streptococcus*
- *Enterococcus*



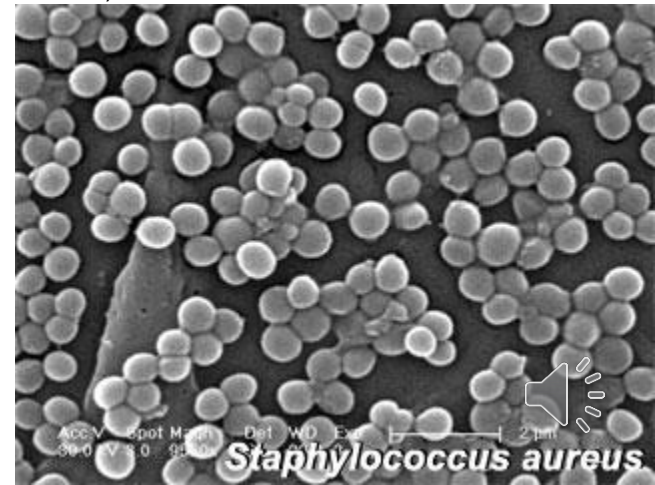
- *Peptococcus*
- *Peptostreptococcus*
- *Sarcina*



Staphylococcus

- Common inhabitant of **skin** and **mucous membranes**
- Nonsporeforming facultative anaerobes
- Non-motile
- Staphyle = a bunch of grapes
- Growth : 15-45 °C
- Growth in 15% NaCl

• Staphylococcus aureus (scanning EM, Image CDC)



Most common *Staphylococcus* spp

Coagulase positive

S. aureus

- anterior nares 50-75% healthy people
- skin & mucous membranes
- hospital environment

Coagulase negative

• *S. epidermidis* & others

- resident skin flora, gut, upper respiratory tract

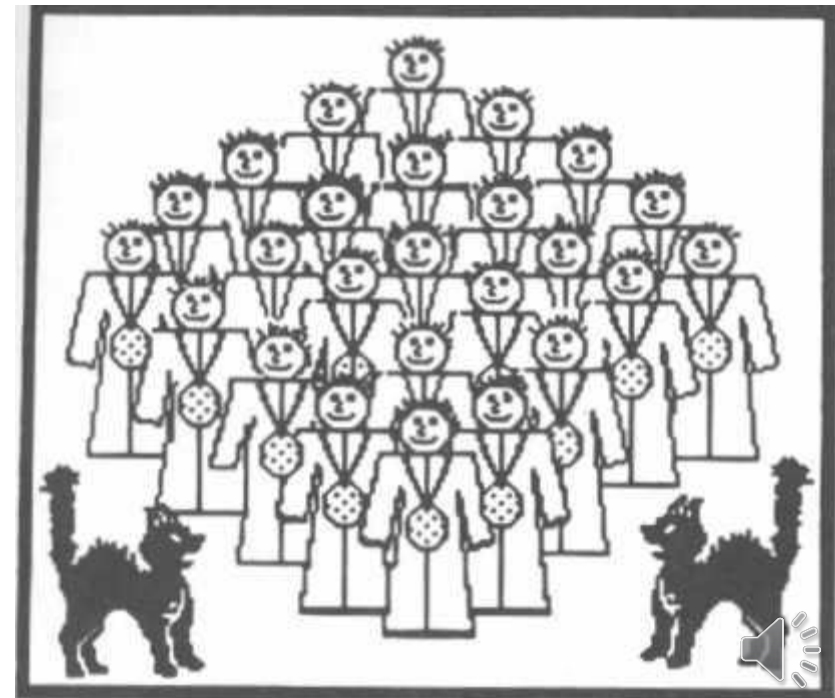
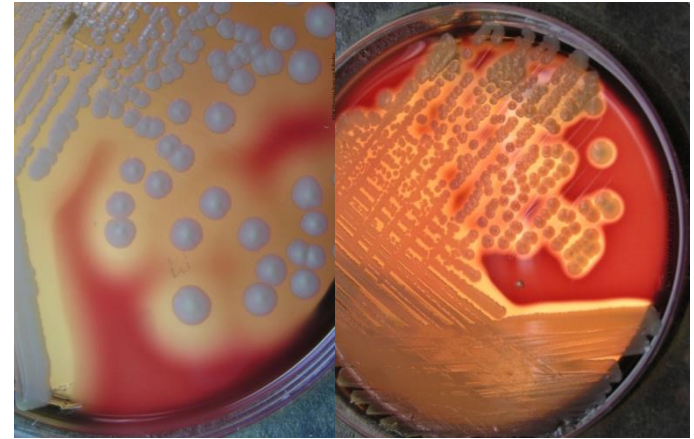
• *S. saprophyticus*

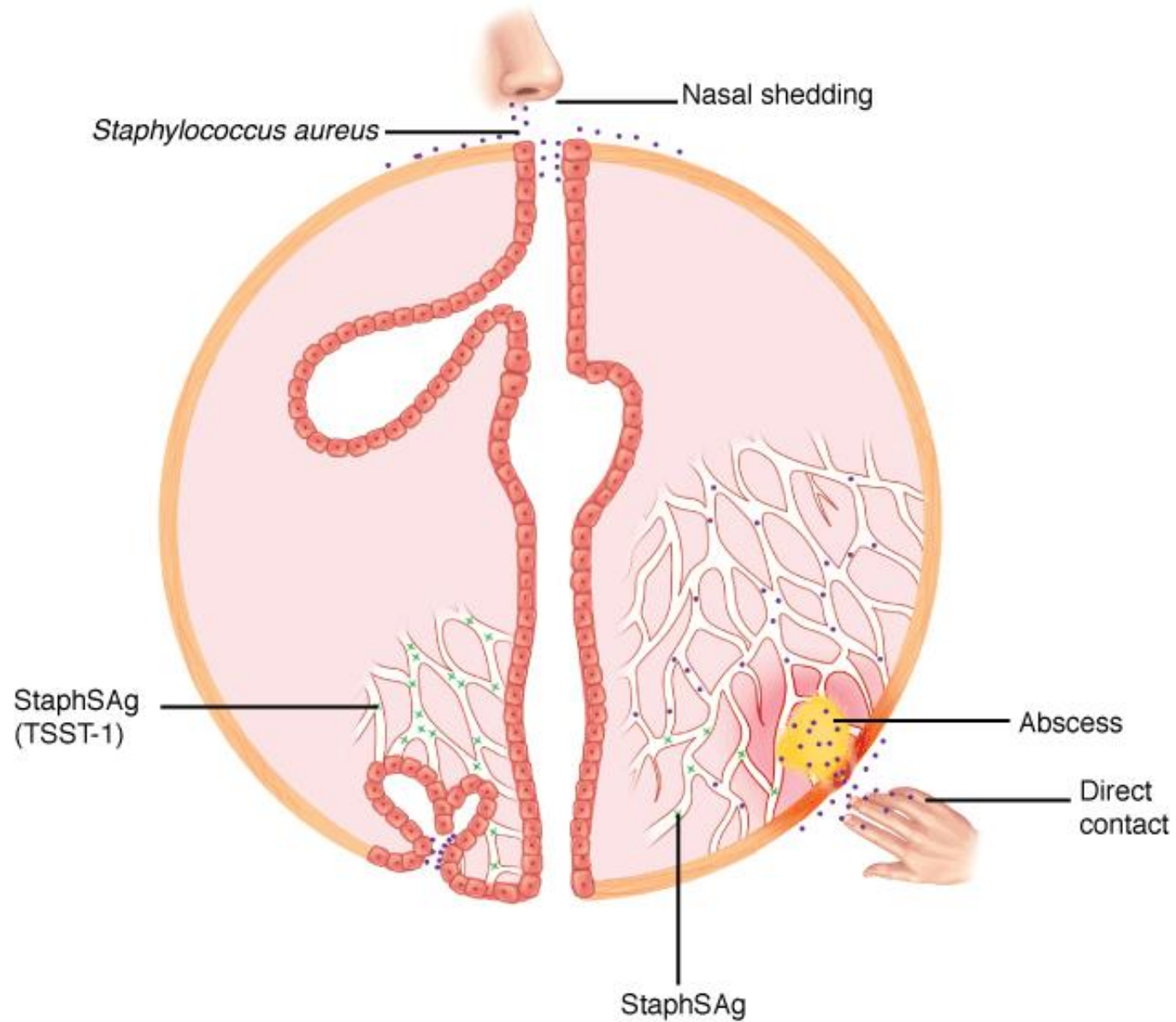
- Urinary tract in young women

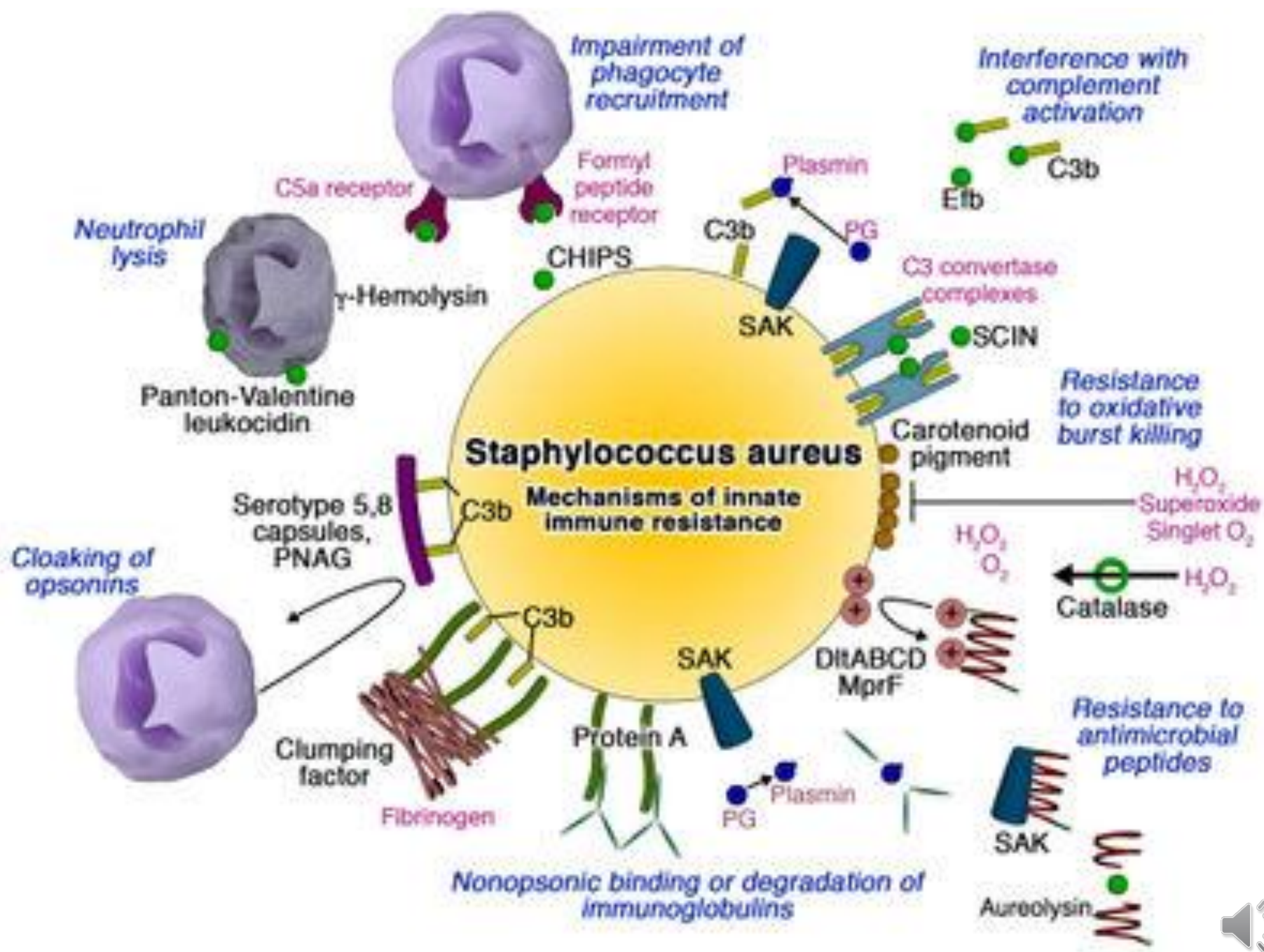


Staphylococcus aureus

- Aureus = gold hence “golden Staph”
- Gram positive cocci
- Colonies pigmented cream to pale yellow
- Usually haemolytic
- Catalase positive
- Coagulase positive
- Fermentation of glucose produces mainly lactic acid
- Ferments mannitol ↗ 1/2 ferment mannitol
(distinguishes from S. epidermidis)
(Cell wall teichoic acid: ribitol teichoic acid)







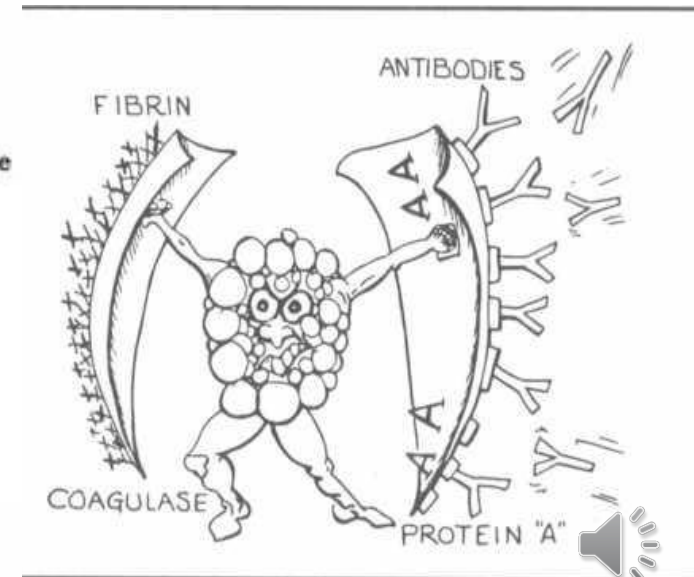
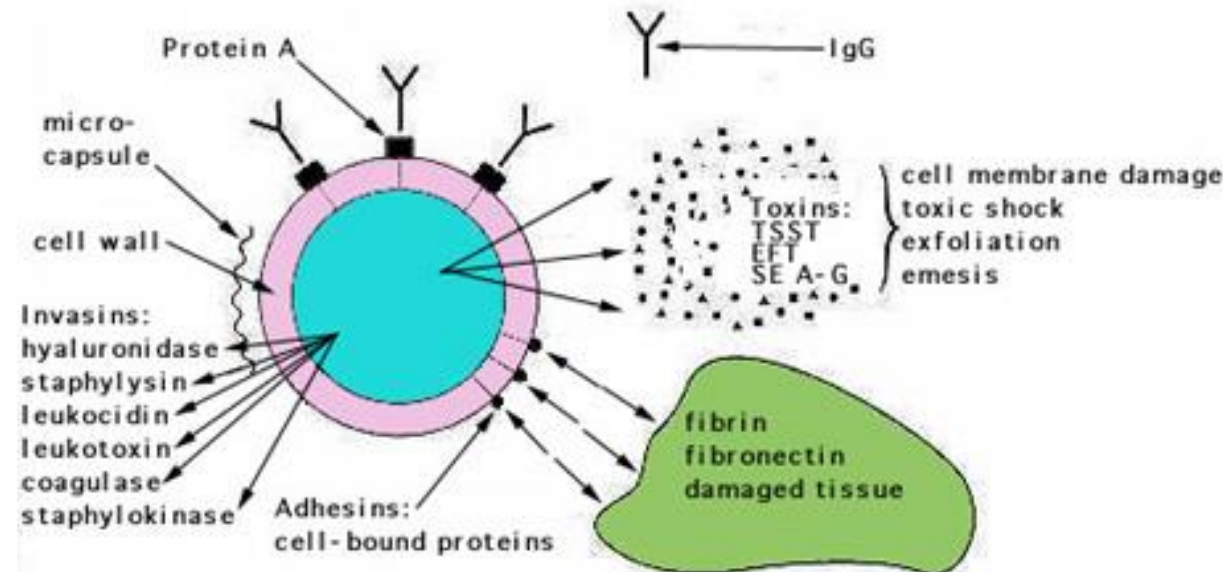
Potential virulence factors

- **Proteins** That Disable Our Immune Defenses
→ ทำให้ immune ใช้งานไม่ได้
- **Proteins** to Tunnel Through Tissue
→ เข้าผ่านอุโมงค์ รั้ว เนื้อเยื่อ
- **Exotoxin** Assault Weaponry
→ เข้าโจมตี ทำลาย เซลล์



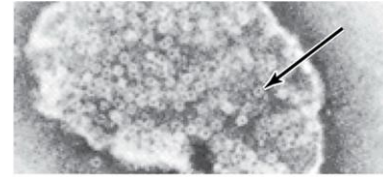
Potential virulence factors

- (1) **surface proteins** that promote colonization of host tissues (fibrin/fibrinogen binding protein (clumping factor))
- (2) **invasins** ^{→ เข้าสู่ within cell} that promote bacterial spread in tissues (leukocidin, staphylokinases, hyaluronidase)
- (3) **surface factors** that inhibit phagocytic engulfment (**capsule**, **Protein A**)
- (4) **biochemical properties** that enhance their survival in phagocytes (**carotenoids**, **catalase** production)

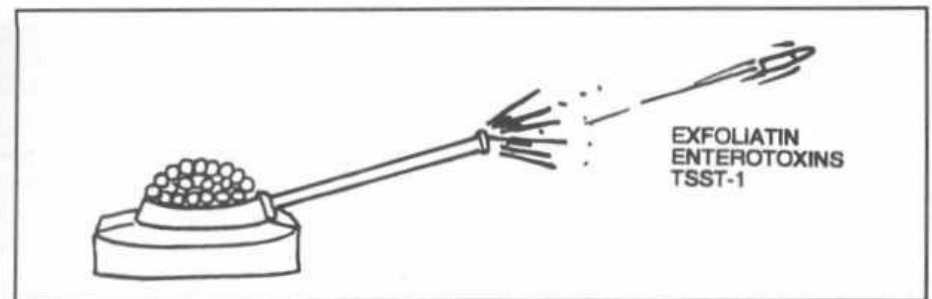


Potential virulence factors

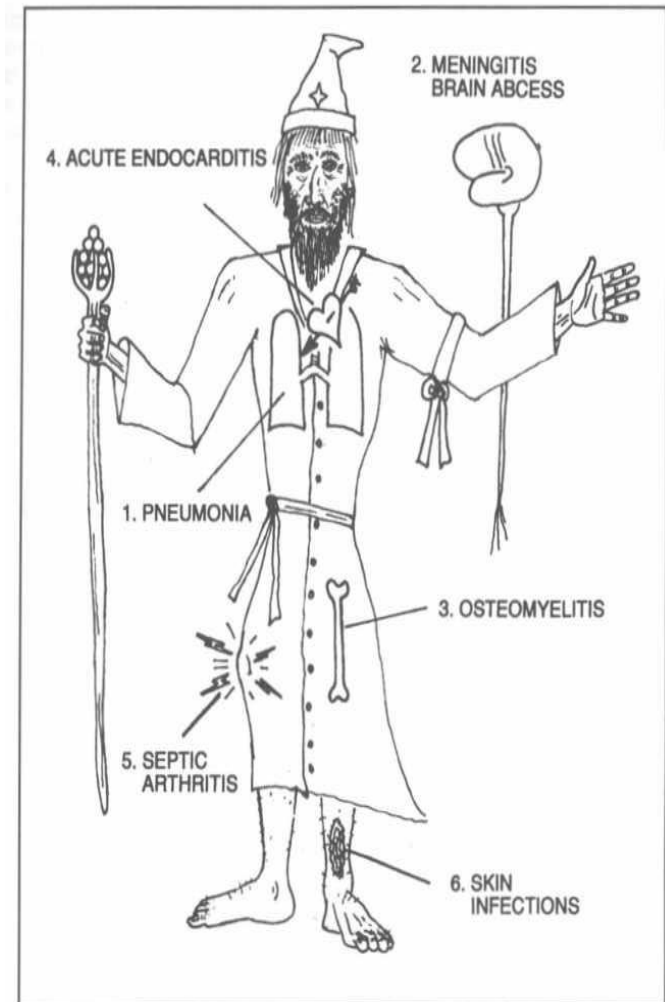
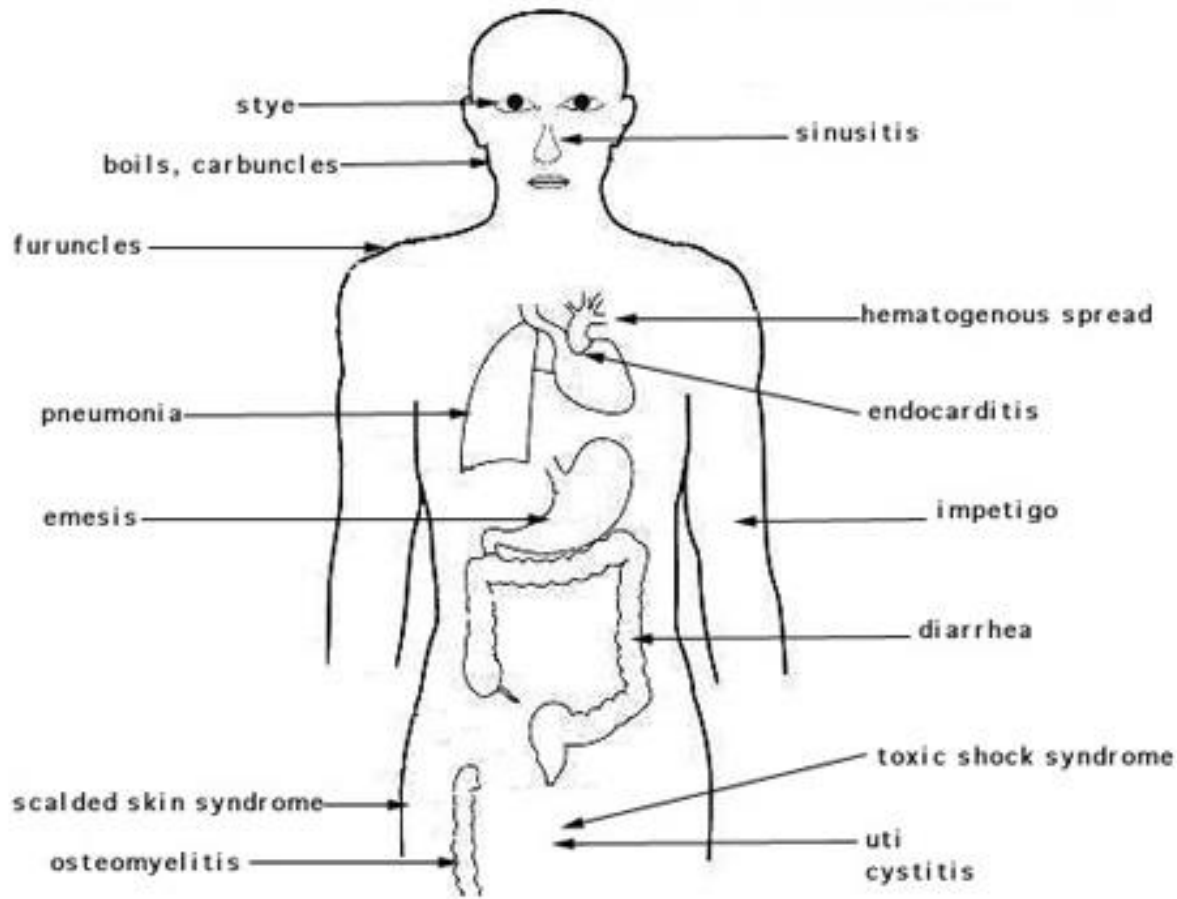
- (5) immunological disguises (**Protein A**, **coagulase**)
- (6) membrane-damaging toxins that lyse eukaryotic cell membranes (**hemolysins**, **leukotoxin**, **leukocidin**, **lipase**, **protease**)
- (7) exotoxins that damage host tissues or otherwise provoke symptoms of disease (**Enterotoxin(SE A-G)**, , **TSST-1** (**superantigen**), **Exfoliative toxin**; **ET-A**, **ET-B** associated with **scalded skin syndrome**)
- (8) inherent and acquired **resistance to antimicrobial agents**.



Source: Ryan KJ, Ray CG: Sherris Medical Microbiology, 5th Edition:
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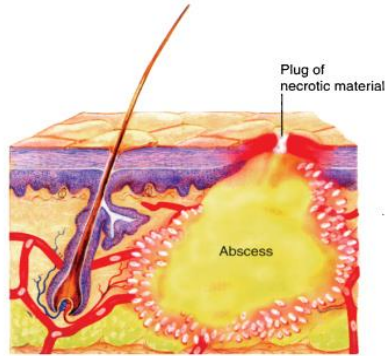
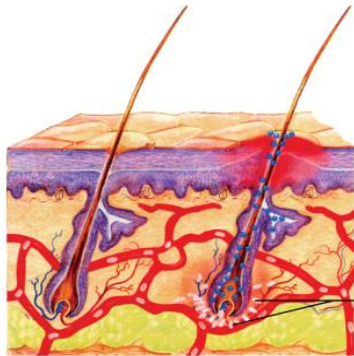
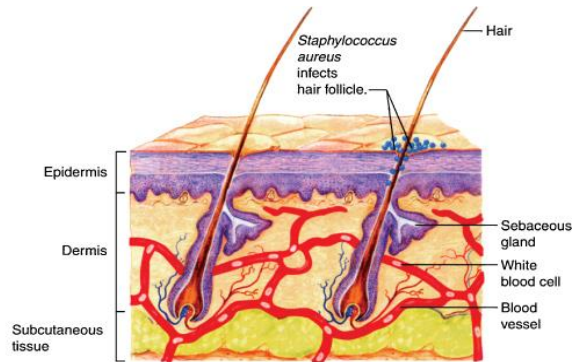


Infection and diseases caused by *Staphylococcus aureus*



๑

Furuncle (boil)



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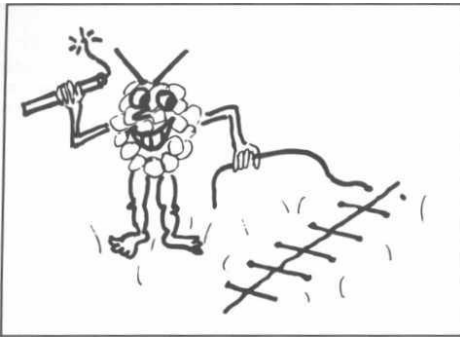
โรคฝีฝักบัว

Carbuncle



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S. aureus Infections



- **Suppurative infection** (Pus forming)

Skin infections: boils, styes, furuncle,
Impetigo, carbuncles, cellulitis
Wound infections/ school sores

- **Systemic infection**

Osteomyelitis/ pyoarthritis, Pneumonia
Bacteraemia/ septicaemia, Meningitis
Endocarditis, UTI

-Hospital acquire - surgical wound
- indwelling devices

- **Food poisoning (food intoxication)**

Enterotoxin (heat stable)



S. aureus Infections

- **Toxin**

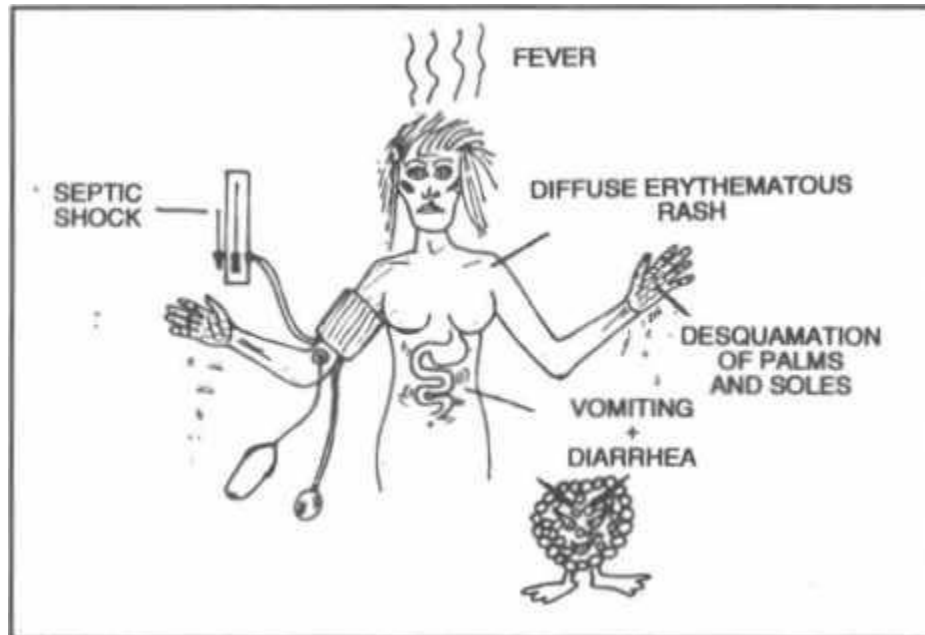
Staphylococcal scalded skin syndrome
(Exfoliative toxin, ET: epidermolytic toxin)



Staphylococcal scalded skin syndrome

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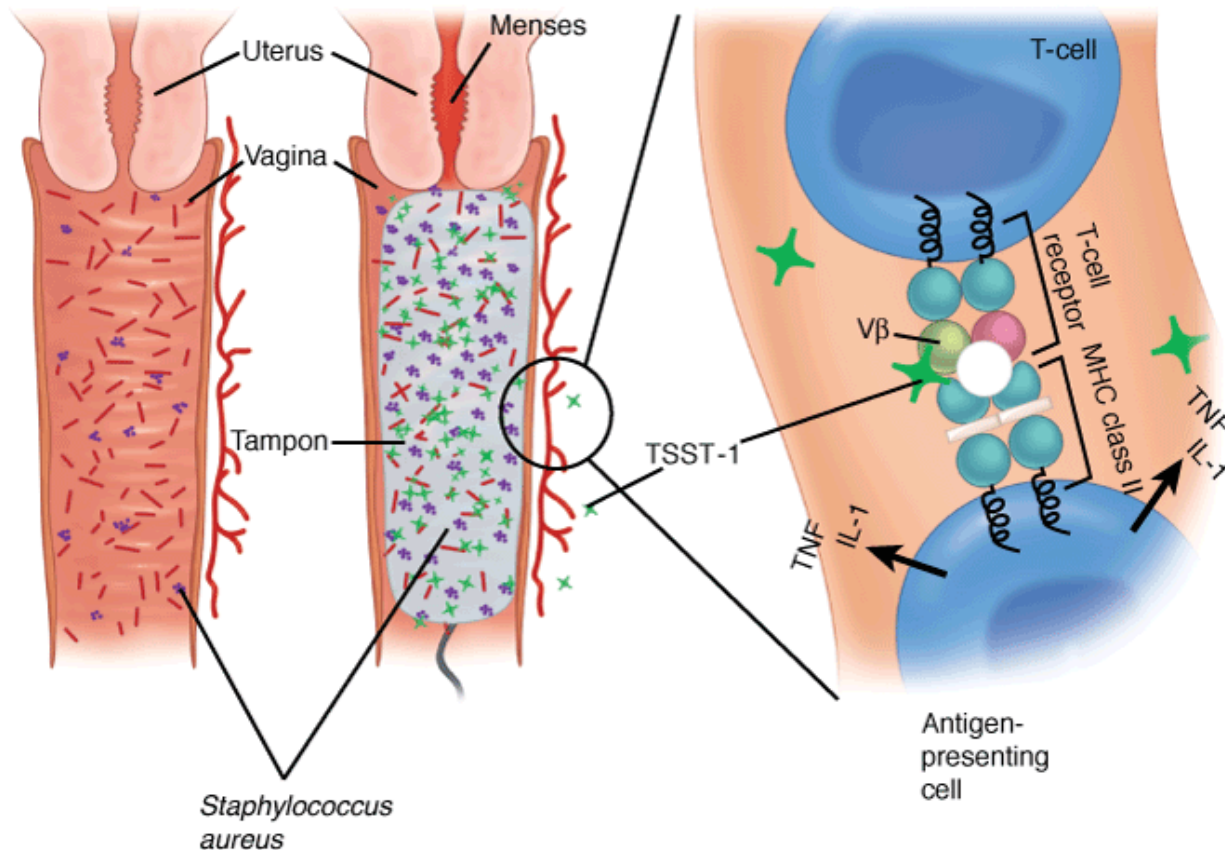
Staphylococcal **toxic shock syndrome**
(multiple organ dysfunction)



A. Vaginal colonization

B. **Toxin** production during menses

C. Toxin absorption, and superantigen stimulation of cytokine production



Staphylococcal toxic shock syndrome
(multiple organ dysfunction) TSST-1 (super antigen)



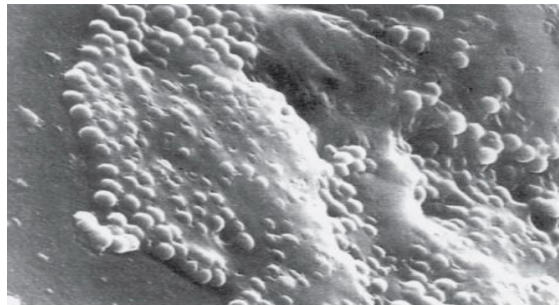
Methcillin resistant *S. aureus* (MRSA)

- Hospital-Associated MRSA (HA-MRSA) or Community-Associated MRSA (CA-MRSA)
- Nosocomial infection
- 95% of *Staph. aureus* resistant to penicillin
- MRSA also resistant to methcillin, oxacillin, gentamicin, cephalosporins, tetracycline, erythromycin
- Treat with vancomycin—VRSA

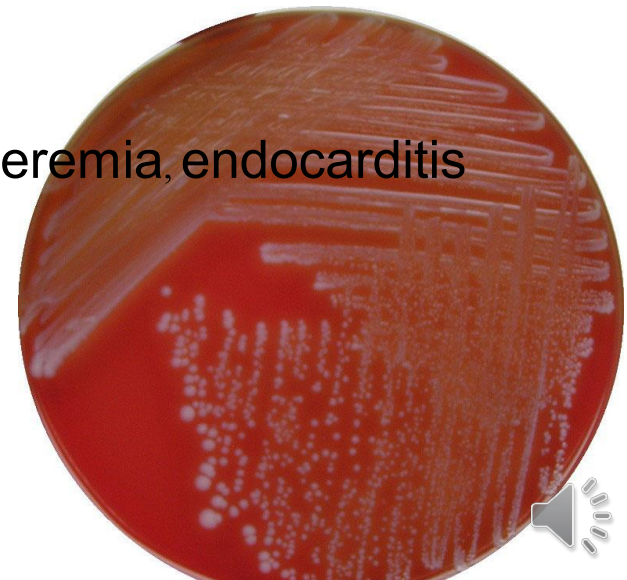


Staphylococcus epidermidis

- Normal flora—skin, mucous membranes
- Usually non-hemolysis
- Produced a slime (secreted teichoic acid) resulting in biofilm formation
- Indwelling devices
- Polysaccharide-adhesin called PS/A
- PIA (polysaccharide intercellular adhesin)
- PGA (poly-gamma-DL-glutamic acid)
- UTI(catheter-associated), Osteomyelitis, Bacteremia, endocarditis etc



^A
Staphylococcal slime



Staphylococcus saprophyticus

រស់នៅ ជាតិ លើស្បែក

- Infrequent resident of skin
- UTI (true pathogen)
- Subacute, chronic
- Young, sexually active females
- Adheres to uroepithelial, urethral and periurethral cell (Ssp, hemagglutinin)
- Urease

ចំណង់ ភ័យ



Staphylococcus lugdunensis

- Common human skin commensal.
→ *สิ่งมีชีวิตที่อาศัยอยู่ร่วมกัน*
- Both community-acquired and nosocomial infections.
- Clinical manifestations: Abscesses, Meningitis, Skin and soft-tissue cellulitis, Prosthetic joint infection, Catheter-related bacteremia, Septic arthritis, Osteomyelitis, Endocarditis, Peritonitis associated with peritoneal dialysis, Toxic shock syndrome.
- Colonies of *S. lugdunensis* are usually hemolytic, sticky, smooth, glossy, yellow, yellow-orange or cream.



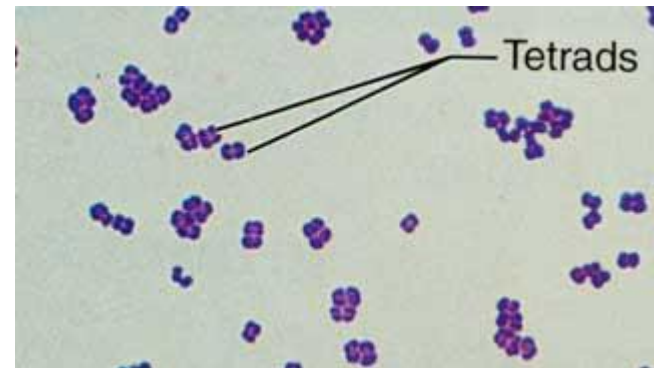
Staphylococcus lugdunensis

- *S. lugdunensis* can usually be successfully identified based on a positive catalase test, negative tube coagulase test, positive PYR test, and a positive ornithine decarboxylase(OD)
- *S. lugdunensis* produces a **bound coagulase** (clumping factor), which is why **the slide coagulase test may be positive!**, however it does not produce free coagulase, and, unlike *S. aureus*, it will always be tube coagulase negative



Micrococcus

- Occurrence in wide range of environments, including water, dust, and soil.
- Skin normal flora, mucous membrane
- 0.5 to 3 micrometers in diameter and are typically appear in tetrads.
- opportunistic pathogen ✖
- *M. luteus*, *M. lylae*, *M. flavus*



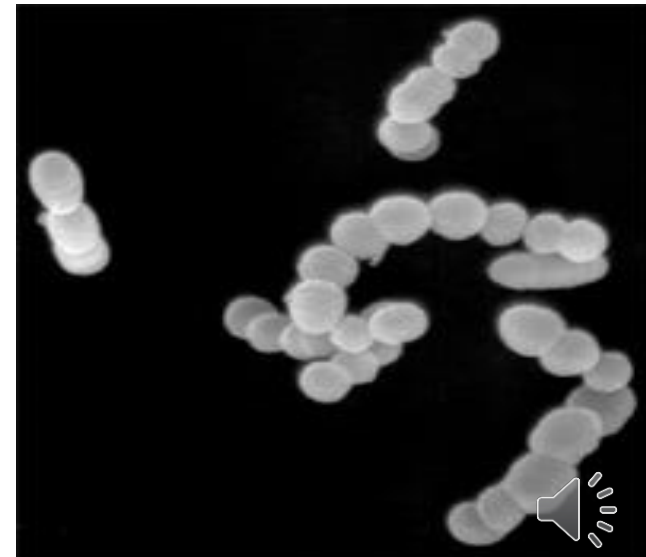
Treatment and Prevention

- Drain abscesses
 - Treat systemic diseases with
 - antibiotics—lengthy
 - Humans principle reservoir
 - Handwashing
 - Dressing disposal, isolation
 - Attention to indwelling devices
- Carriers—ID, restrict, treat



Streptococcus

- Facultative anaerobes, Homofermentative
- Some normal flora, some pathogens
- Sensitive to heat, drying, etc.
- Do not readily develop drug resistance
- Also identified by hemolysis and Lancefield groups
- Catalase negative
- Non-motile, non-spore-forming,



Streptococci classification

- Hemolytic pattern
- Physiologic characteristic
Pyogenic streptococci, Lactococci,
Enterococci, Viridans streptococci
- Serologic classification of
Lancefield.
- Biochemical characteristic
- Genetic



Serologic classification of Lancefield

- Rebecca Lancefield (from A through V)
- β -hemolytic
- Specific cell wall polysaccharide (C-carbohydrate)

Strep. Group A ramnose-N-acetylglucosamine

Strep. Group B ramnose-glucosamine

Strep. Group C ramnose-N-acetylgalactosamine

Strep. Group D glycerolteicoic acid

- In human infection: group A, B, C, D, F, G and N



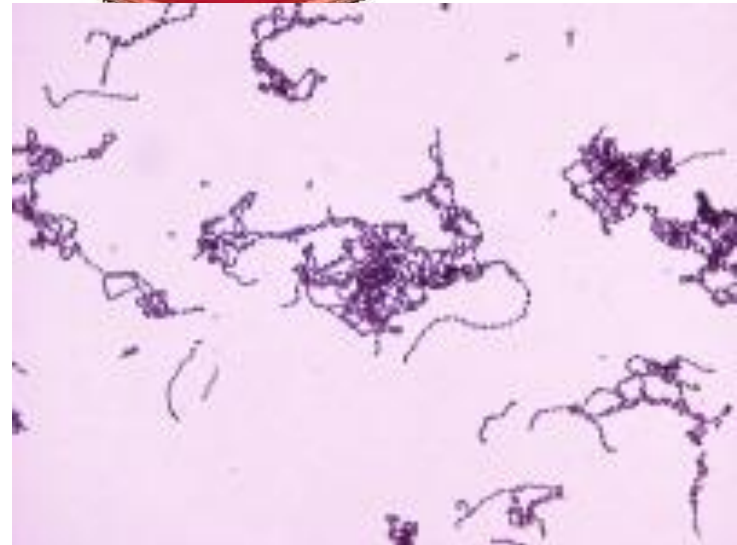
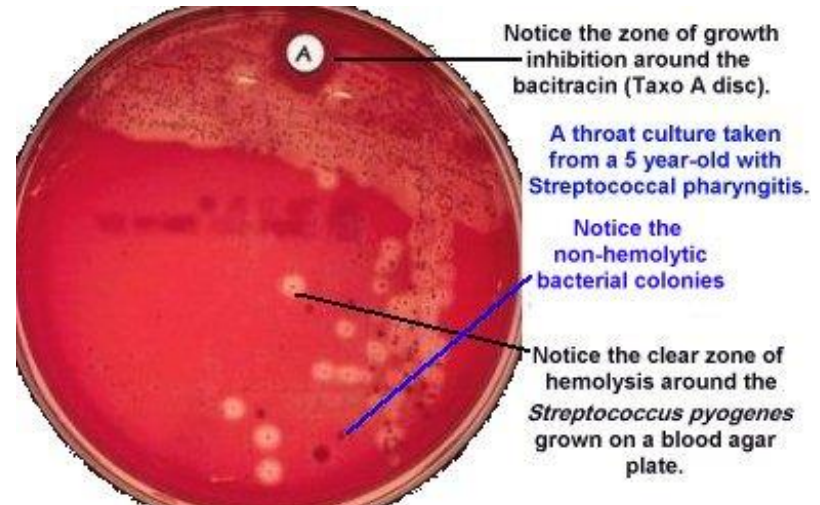
Serologic classification of Lancefield.

- Group A Strep--*Streptococcus pyogenes*
- Group B Strep--*Streptococcus agalactiae*
- Group D Strep--*Streptococcus bovis*,
Enterococcus (Streptococcus) faecalis

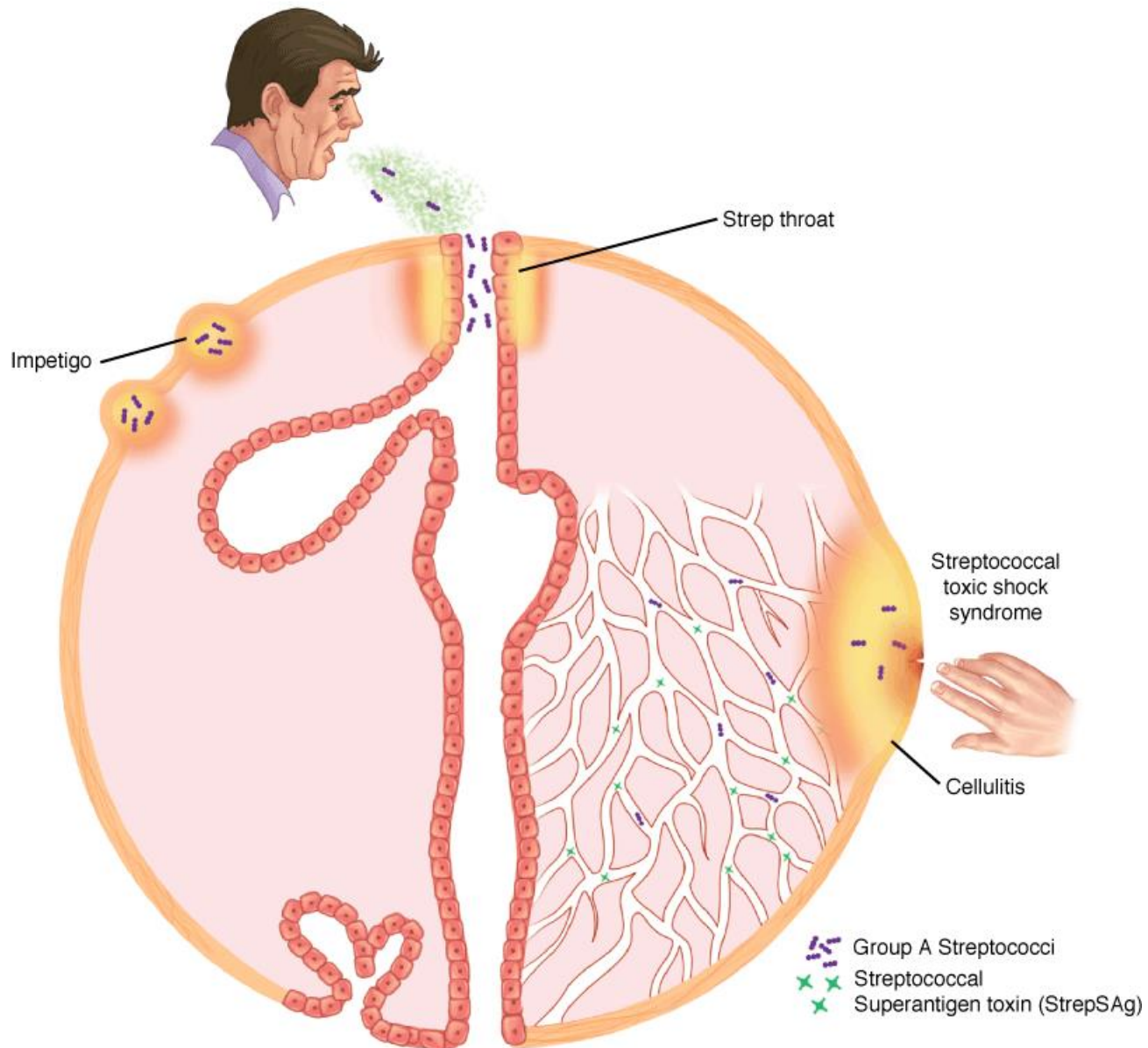


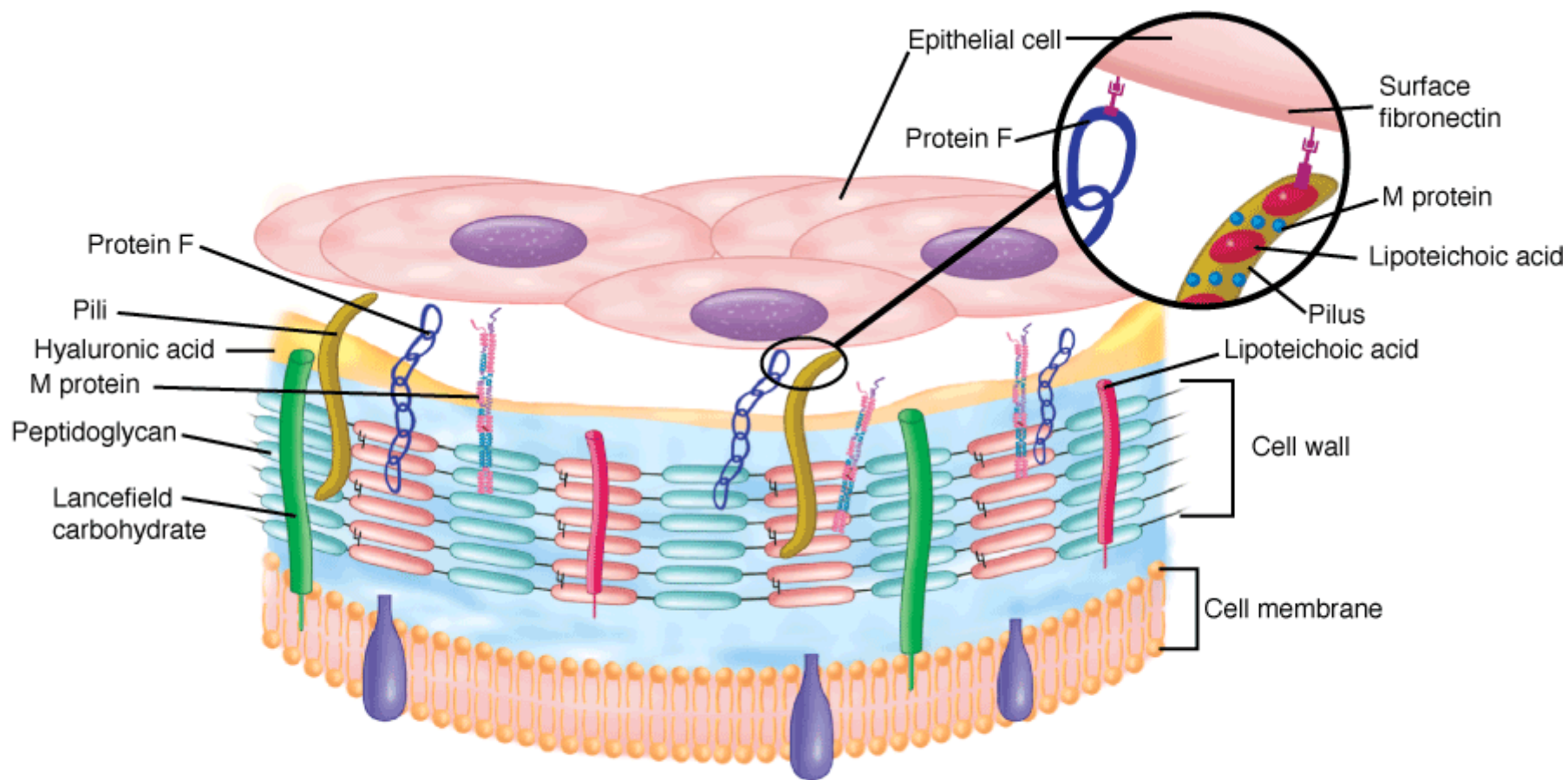
Streptococcus pyogenes

- β -hemolytic
- Group A
- Humans only
reservoir—carriers
- Most serious
pathogen of genus
(90%)
- Strict parasite—
inhabits throat,
nasopharynx, and
skin



Group A Streptococcus





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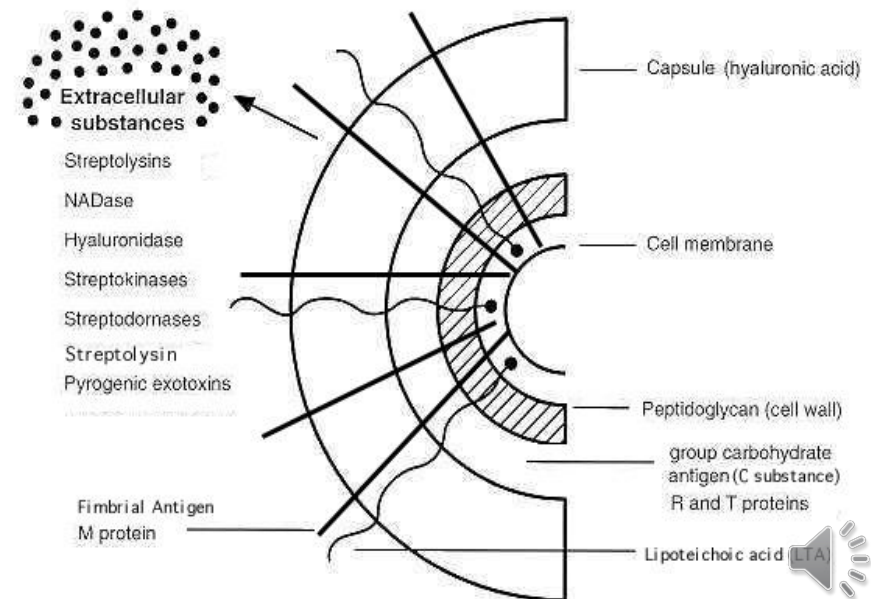
Virulence factor

- (1) M protein, fibronectin-binding protein (Protein F) and lipoteichoic acid for adherence
- (2) hyaluronic acid capsule as an immunological disguise and to inhibit phagocytosis; M-protein to inhibit phagocytosis
- (3) invasins such as streptokinase, streptodornase (DNase B), hyaluronidase, and streptolysins; (streptolysin O, streptolysin S)
- (4) exotoxins, such as pyrogenic (erythrogenic) toxin which causes the rash of scarlet fever and systemic toxic shock syndrome.



Source: Ryan KJ, Ray CG: *Sherris Medical Microbiology*, 5th Edition:
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Group A Strep and Infections

1) ^{suppurative} Suppurative

Upper respiratory tract infection (URI)

Strep' throat – pharyngitis (pyrogenic exotoxin:

Scarlet fever (red rash), strawberry tongue), tonsillitis,

Sinusitis

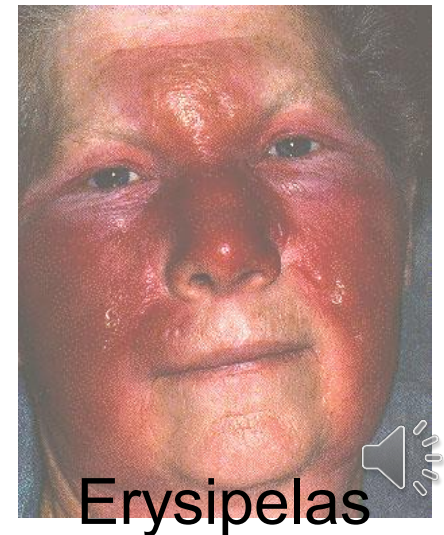
Skin Infection: pyoderma, Impetigo (superficial)

2) Systemic/ deep tissue

^{โรคไฟลามทุ่ง}
Erysipelas, Cellulitis (deep), necrotizing fasciitis

(flesh-eating bacteria), systemic toxic shock syndrome

3) Non-suppurative sequelae

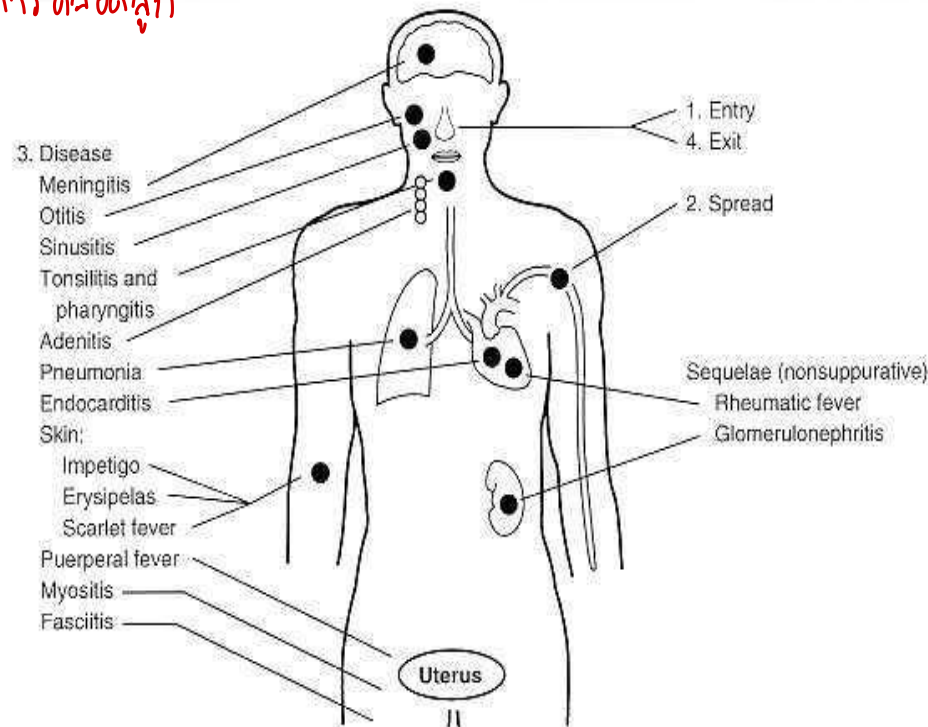


Erysipelas

Group A Streps and Infections

- Streptococcal toxic shock syndrome
(streptococcal pyrogenic exotoxin (Spe))
- Puerperal fever (postpartum endometritis)

הזיהום הנגרם על ידי



Non-suppurative sequelae

ภาวะแทรกซ้อนแบบไม่ขึ้นแผล

Only Streptococcus pyogenes

กรวยไตอักเสบ Glomerulonephritis or Goodpasture's syndrome

- Follows pharyngitis or skin infections
- Inflammation of the renal glomeruli
- Deposition of streptococcal Ag–Ab–C' complexes on the renal glomeruli basement membranes

Rheumatic fever โรคไข้วมาติก

- Follows pharyngitis
- Antibodies against a streptococcal cell wall component (M-protein) cross-react with antigens on cardiac myofiber protein myosin

Autoimmune cross-reaction: Molecular mimicry

Anti-Streptolysin O (ASO)
Anti-Dnase B (A-DNaseB)



Streptococcus agalactiae

β-hemolytic

Group B (Baby)

May colonize vagina, pharynx,
and large intestine

Infections

- Neonatal meningitis
- Screen in third trimester

Enzymes: Hyaluronidase,
neuraminidase, protease,
hemolysin, Dnase.

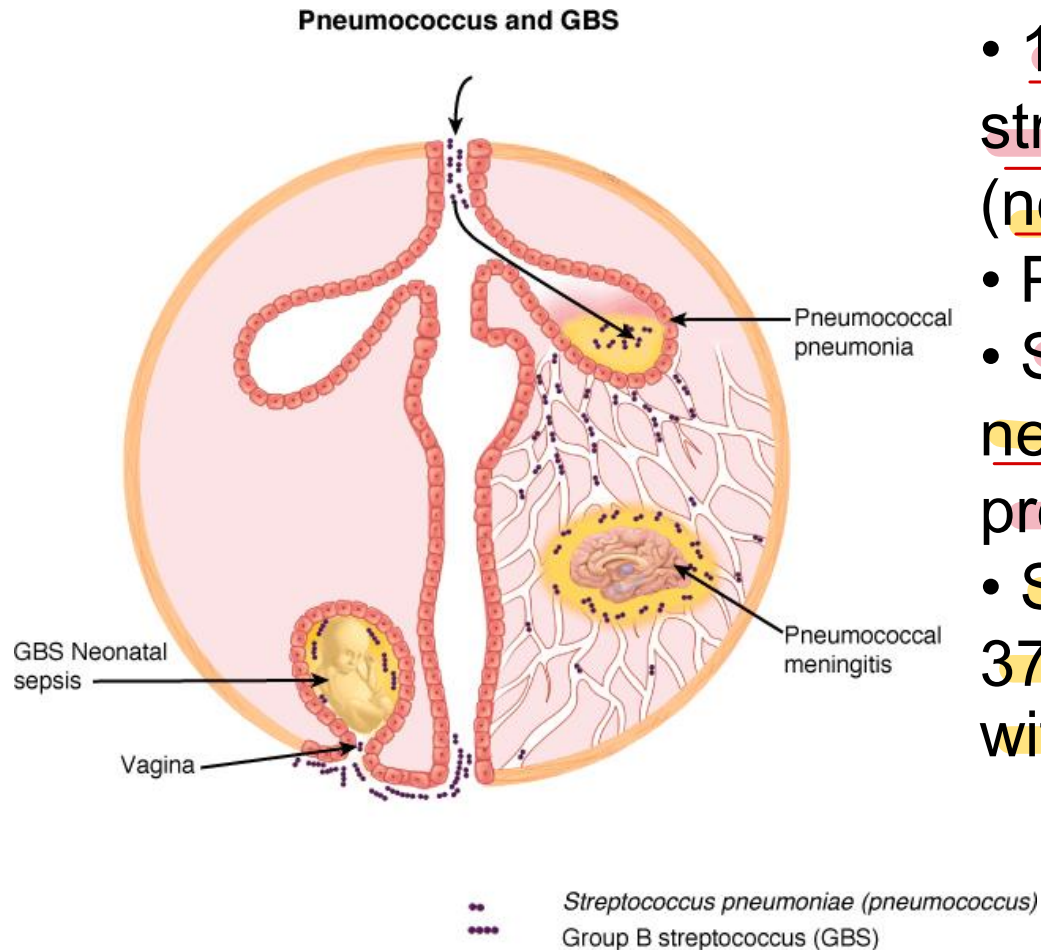


ASM MicrobeLibrary.org © Buxton



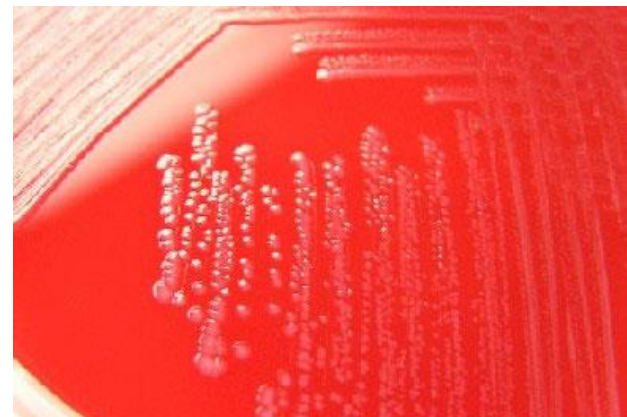
Group B Strep and Infections

- Neonatal infections
- 1 in 4 women carry group B strep in vagina (normal flora)
- Passed to baby during delivery
- Septicaemia and meningitis in newborn – more severe in premature babies
- Screen pregnant women at 35-37 weeks gestation and treat with antibiotics

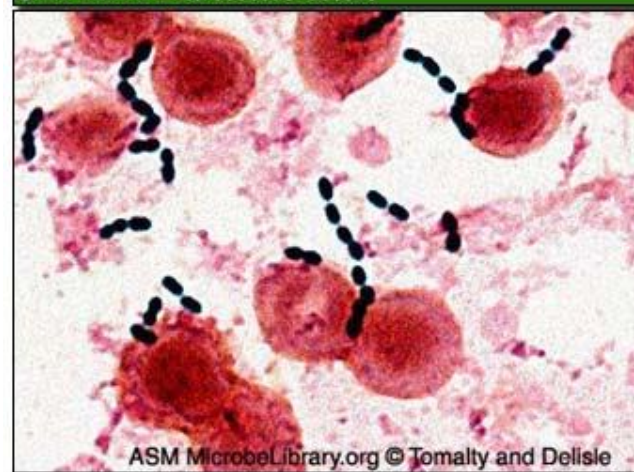


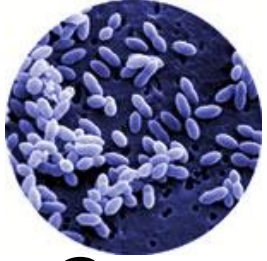
Enterococcus spp

- Normal flora in gut
- 2 species may cause infections, usually hospital acquired, commonly UTI
 - *E. faecalis*
 - *E. faecium*
- Vancomycin resistance *
becoming a problem
(VRE)



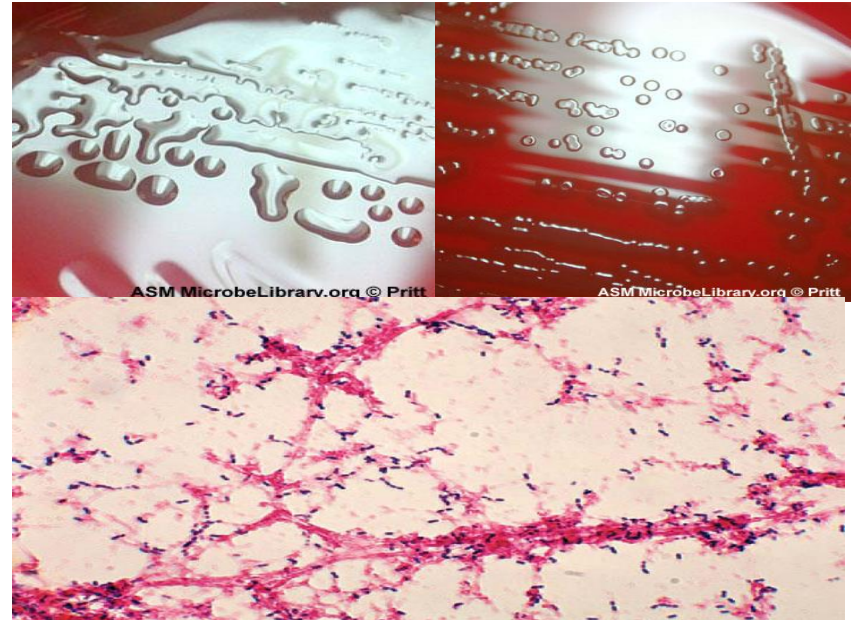
Gram stain of *Enterococcus faecalis*



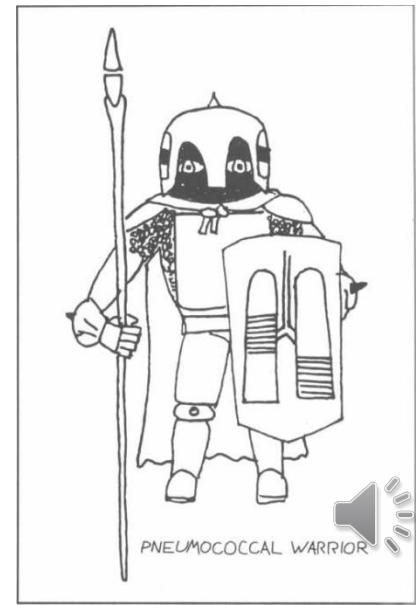


Streptococcus pneumoniae

- Gram positive coccus
- “Lancet-shaped diplococci”
- α -hemolytic
- Fastidious bacterium
- Fermentative aerotolerant anaerobe
- Sensitive to optochin
- Catalase negative
- Mucoid colonies due to large capsule $\rightarrow$ *mucoid*
- (Serological typing)



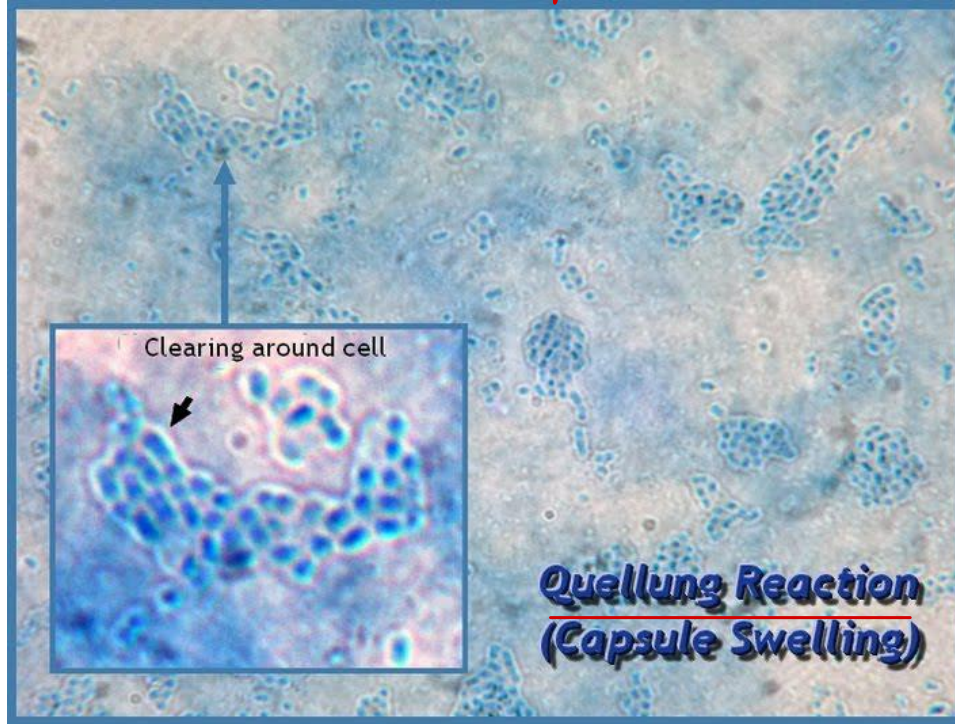
Lancet



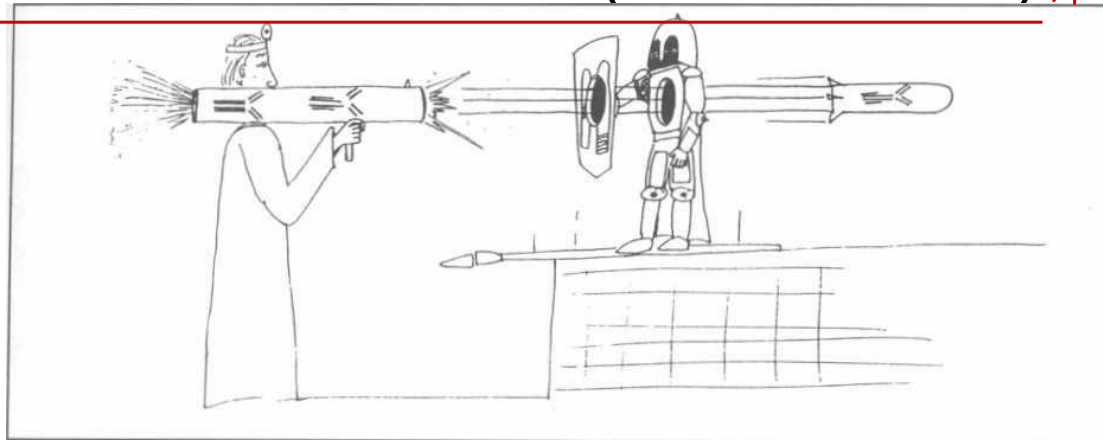
or streptococcus

pneumoniae

Capsule swelling test



* Pneumococcal vaccine(Pneumovax)*



Self-Autolysis

Disrupt and disintegrate itself
by Autolysin *

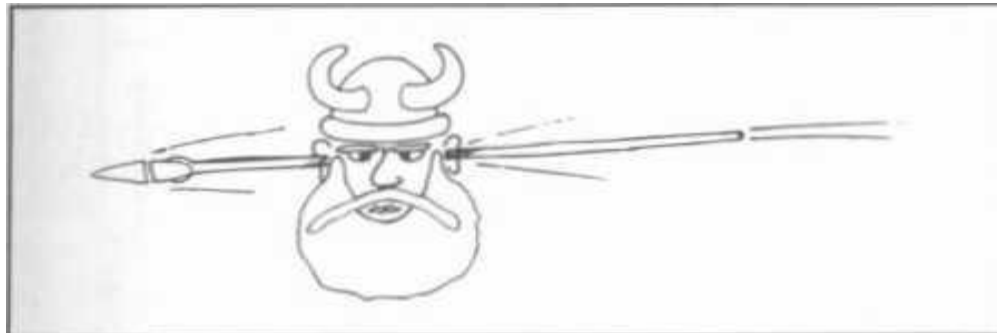
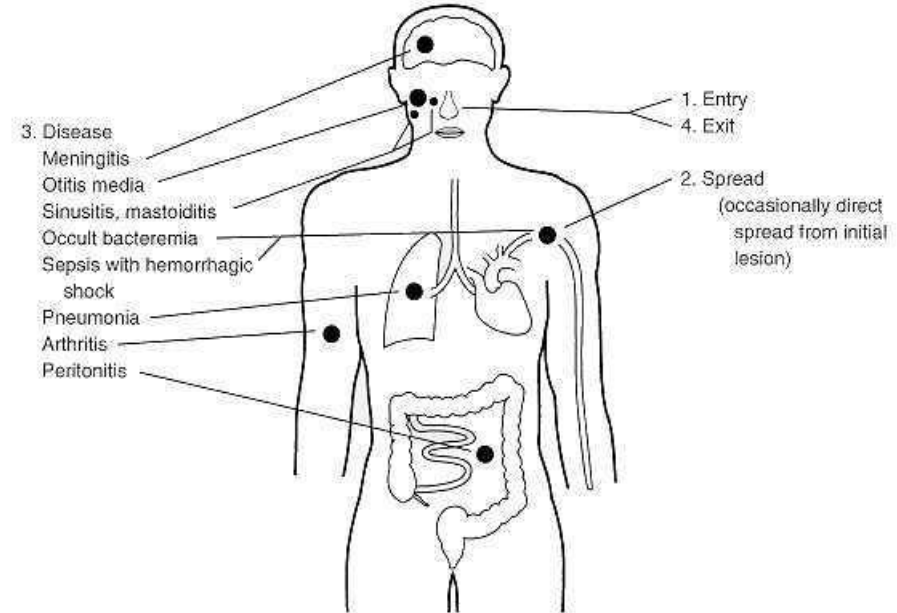
Autolysis usually begins
within 18-24 hours



S. pneumoniae Infections

- **Pneumonia**
- **Septicaemia and Meningitis**
- Soft tissue infections
 - Peri-orbital cellulitis
- Sinusitis
- **Otitis media**
- **Most prevalent in young children (< 5 yrs) and elderly (> 65 yrs)**

อาจได้ทั้งไปและติด



Virulence factors

- Pili: contribute to colonization of upper respiratory tract
- Capsule: interference with ^{immunity} binding of complement C3b to the cell surface
- Choline Binding Proteins:
 - PspA (protective antigen); inhibit complement-mediated opsonization
 - LytA, B, and C (three autolysins)
 - CbpA (an adhesin).
- Pneumolysin
- Neuraminidases, IgA protease



Viridans Streptococci

α -hemolytic; **viridans**-----**VERDE** (green)

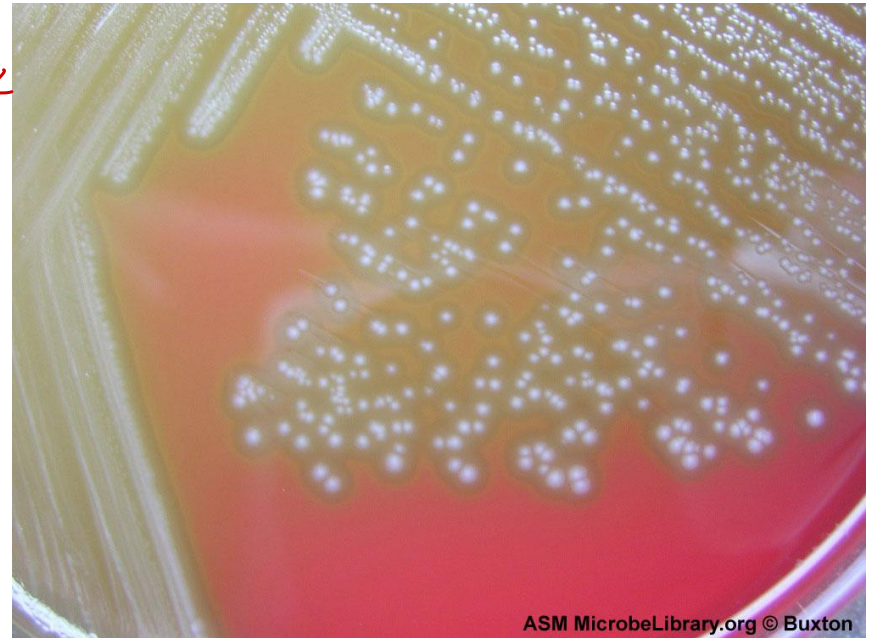
No Lancefield carbohydrates

Normal oral flora

* Opportunistic pathogens *

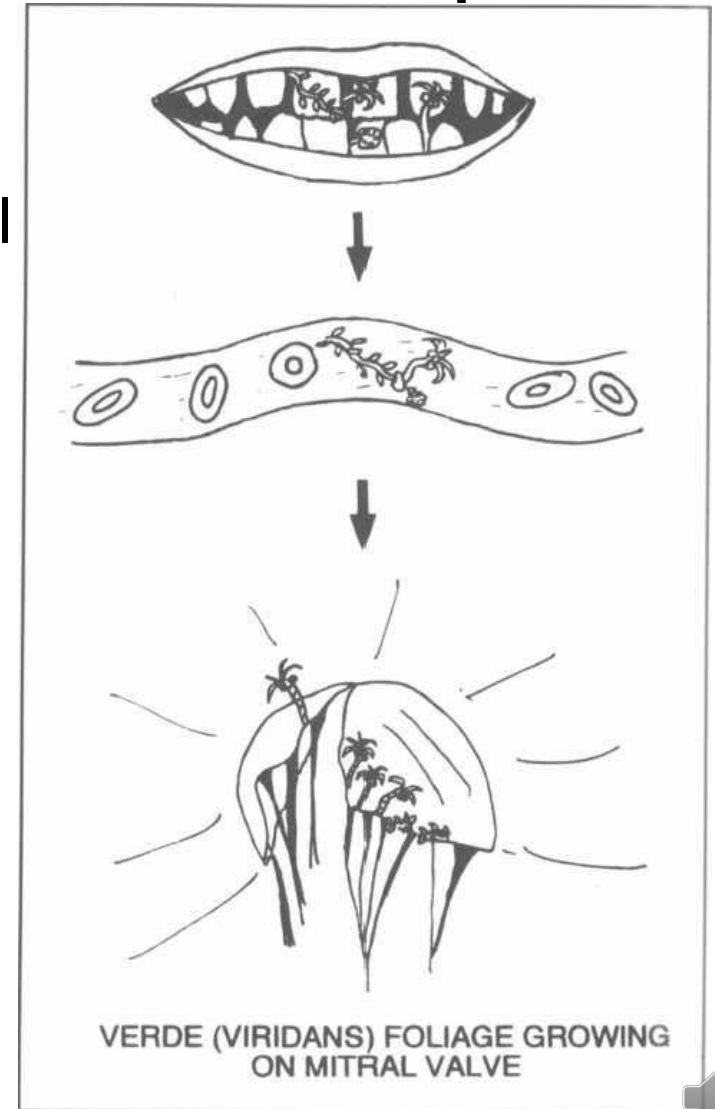
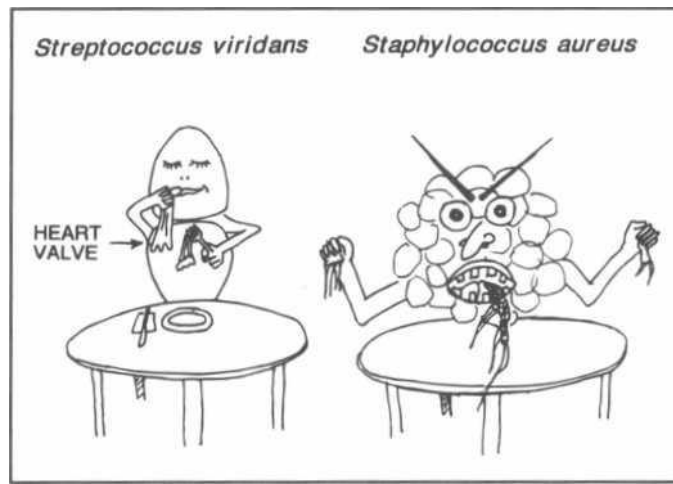
Common species

- Strep. mitis
- Strep. mutans
- Strep. milleri
- Strep. salivarius
- Strep. Sanguis
- Strep. Suis



Pathogenesis of Viridans Strep.

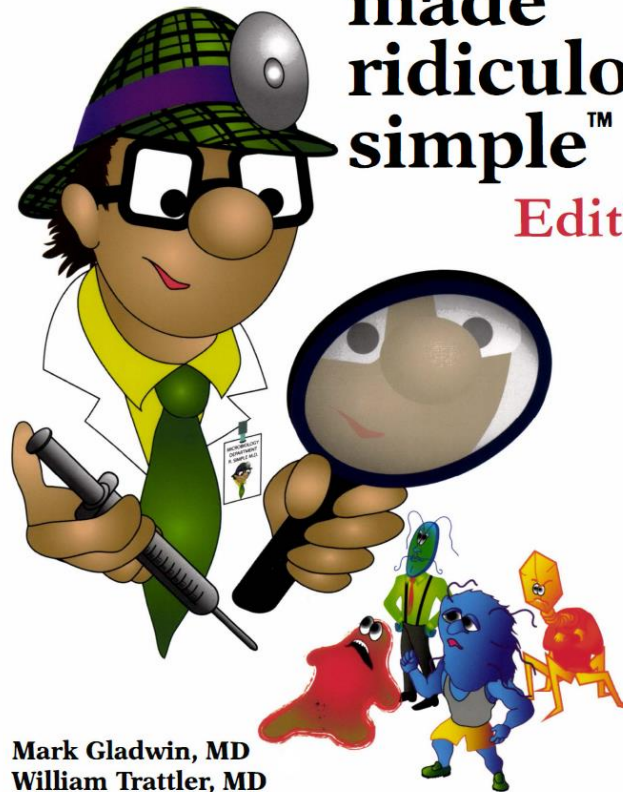
- Not invasive
- Enter blood stream through dental or oral surgery (dental infection)
เข้ากระแสเลือดผ่านช่องปาก
- SBE—Subacute Bacterial Endocarditis (biofilm on valves)
- Abscesses
- Dental plaque and dental carries



CLINICAL MICROBIOLOGY

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simple™

Edition 6



Mark Gladwin, MD
William Trattler, MD
C. Scott Mahan, MD



กินหมูดิบ ระวังหูดับ

กับ โรคติดเชื้อสเตร็ปโตค็อกคัส ซูอิส



ชาวบ้านภาคเหนือมักเรียกกันติดปากว่าโรคหูดับ เพราะผู้ป่วยเมื่อหายแล้วส่วนใหญ่มักมีอาการหูหนวก ไปตลอดชีวิต โรคนี้เป็นโรคติดต่อจากสัตว์สู่คนที่สำคัญโรคหนึ่งที่มีความรุนแรง นอกจากจะทำให้เสียชีวิตแล้ว ผู้ที่รอดตายอาจมีความพิการเหลืออยู่

สนใจรายละเอียดเพิ่มเติม

<http://thaigcd.ddc.moph.go.th>

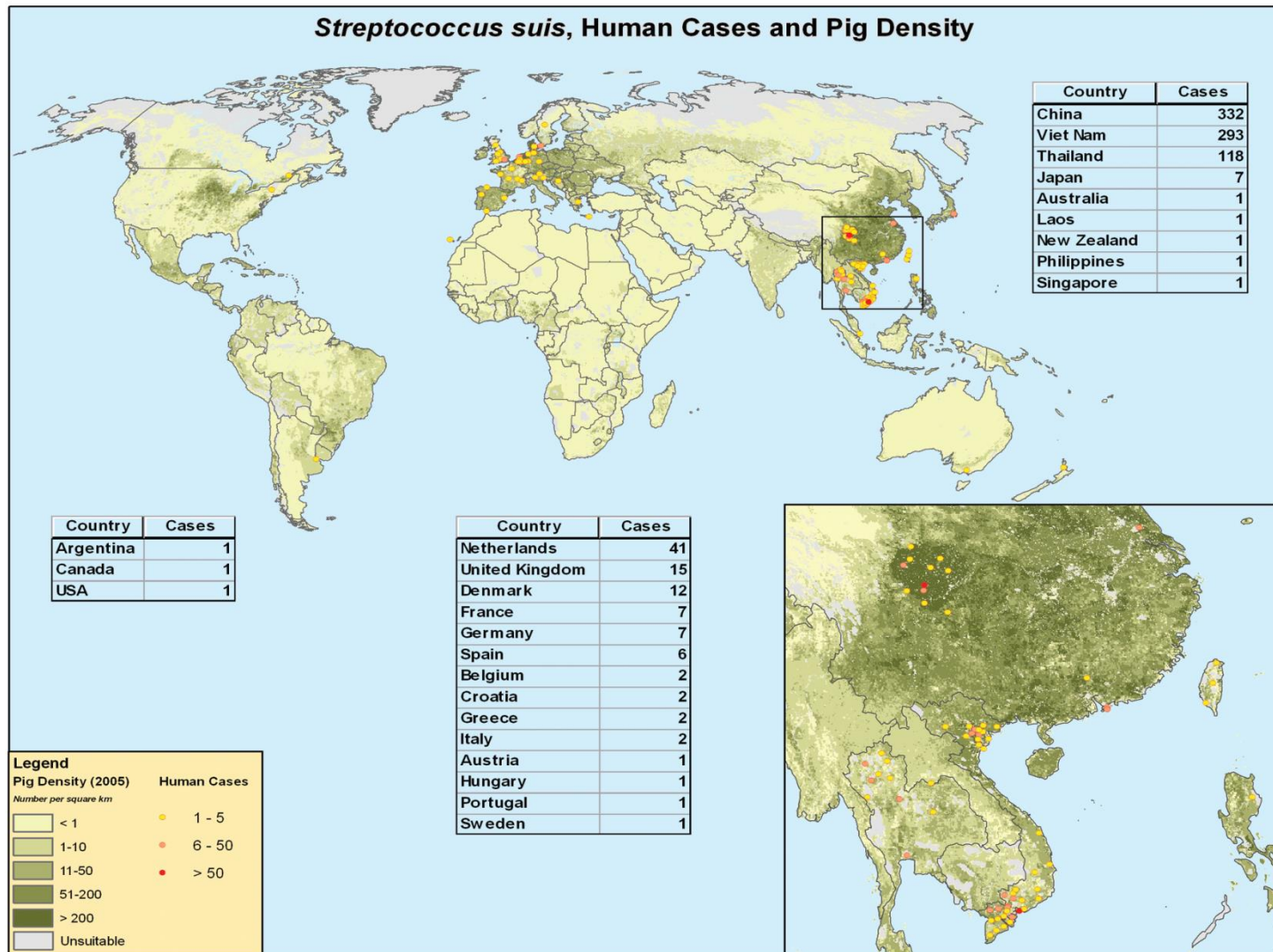


สำนักโรคติดต่อทั่วไป

กรมควบคุมโรค กระทรวงสาธารณสุข



World map of human *Streptococcus suis* cases



Streptococcus suis

- Swine industries
- Pathogen in pigs
- Severe systemic infection in humans
-*S. suis* serotype 2
- Endocarditis, bacteremia, meningitis, hearing loss, acute pyogenic arthritis, endophthalmitis,
- α -hemolytic
- Capsular serotype
- Lancefield group R, S, T and RS

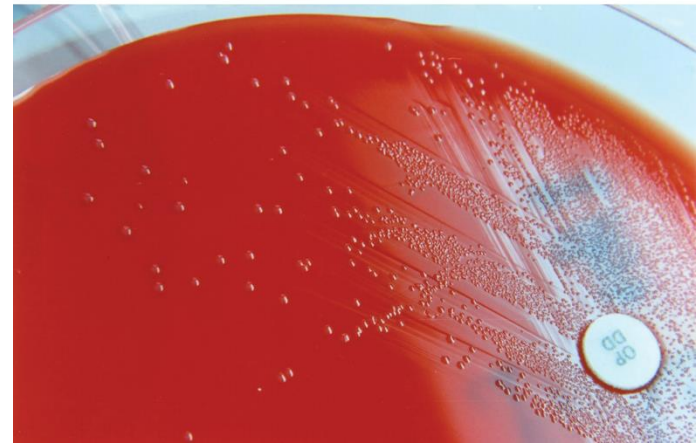
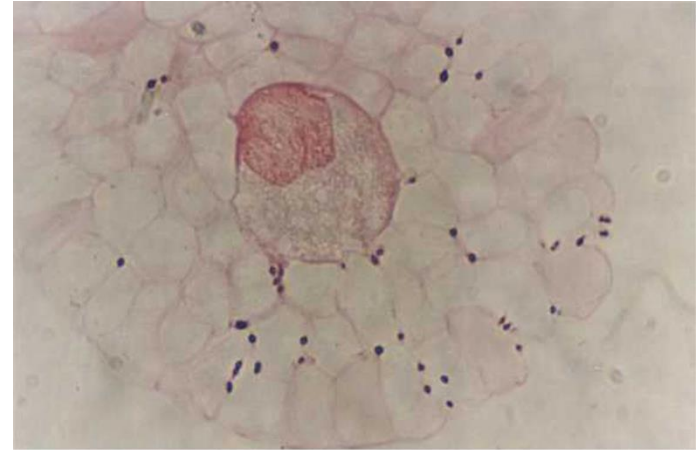


purpura fulminans

Streptococcus suis

Virulence factors

- capsular polysaccharide,
- extracellular protein factor
- muramidase-released protein
- suilysin
- several adhesins
- hyaluronate lyase



Streptococcus suis

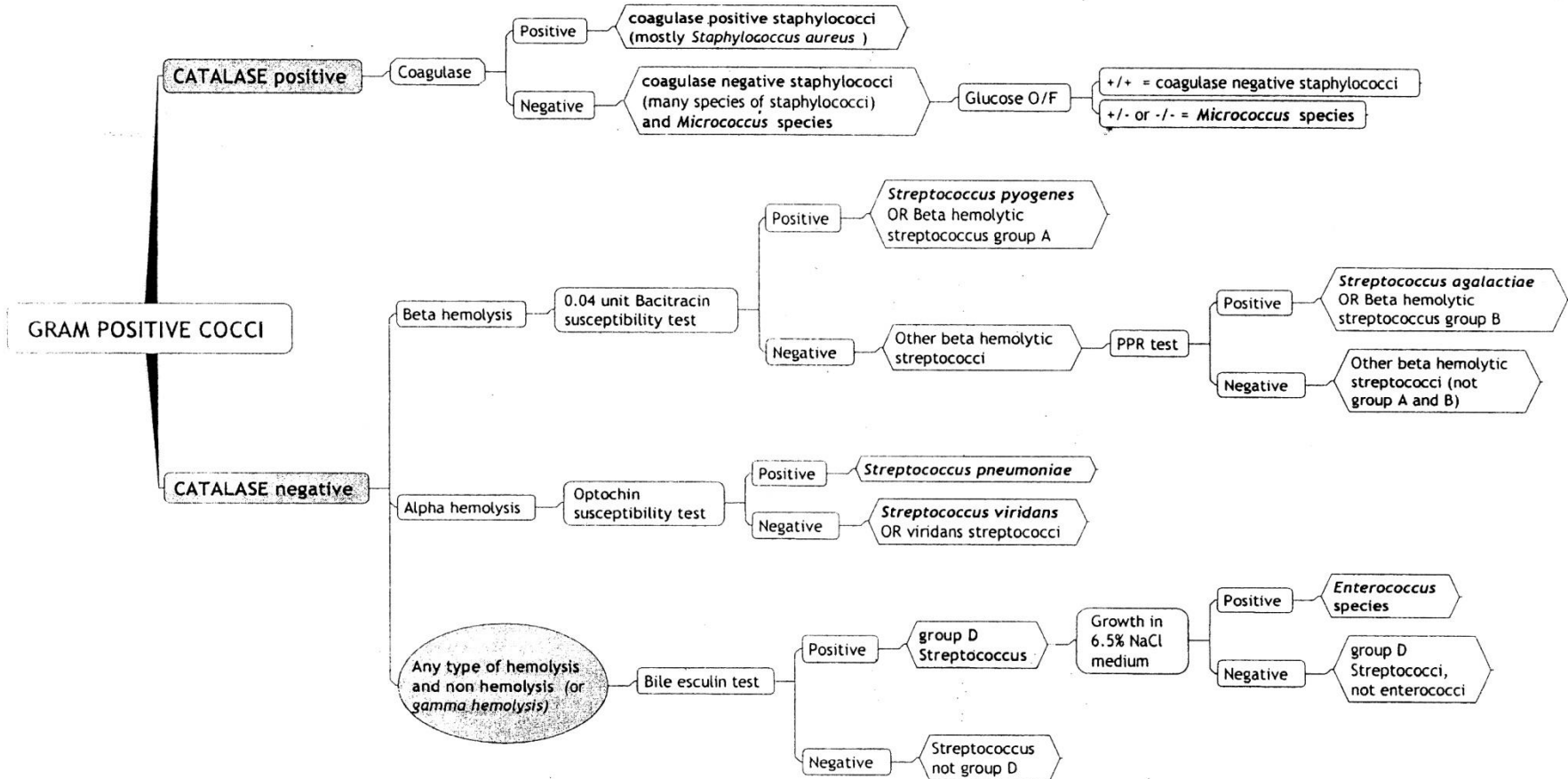
Biochemical Parameters

- Voges-Proskauer negativity
- hydrolysis of esculin positivity,
- trehalose positivity,
- negativity for growth in 6.5% NaCl,
- absence of beta-hemolysis on sheep blood agar.
- *S. suis* serotype 2 is negative for hippurate, pyrrolidonylarylamidase, and mannose.

PCR: *cps2J* gene



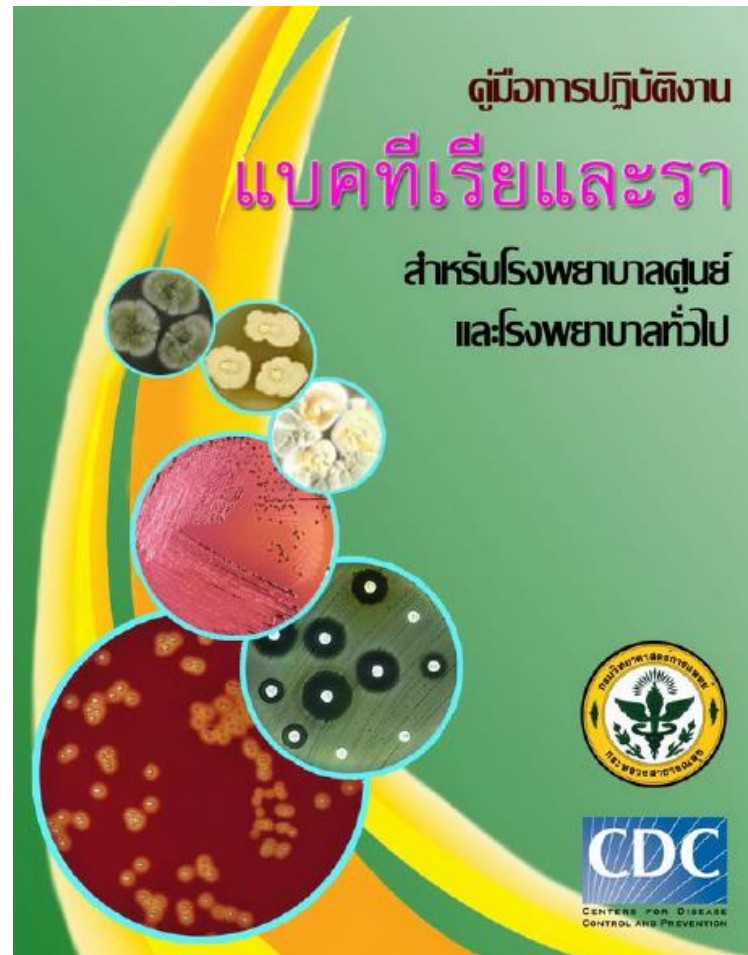
IDENTIFICATION OF GRAM POSITIVE COCCI



* Hemolysis = hemolysis on sheep blood agar plate

**Beta hemolytic streptococcus not group A, B, D = Beta hemolytic streptococci with negative bacitracin susceptibility test, negative PPR test and negative bile esculin test





<http://narst.dmsc.moph.go.th/data/Idenbook.pdf>



ตารางที่ 5.1-1 การวินิจฉัยเชื้อ gram-positive cocci, catalase บวก ที่มีโคโลนีขนาดใหญ่สีขาวหรือเหลือง
ที่พบบ่อย

Organisms	คุณสมบัติเฉพาะ	Slide coagulase	Tube coagulase	Bacitracin (0.04 U)	Polymycin B (300 U)	PYR
<i>Micrococcus</i> group ^a	often yellow			S	S	V
<i>Staphylococcus aureus</i>	VP +	V	+	R	R	-
<i>S. intermedius</i> (dogs)	VP -	V	+	R	S	+
<i>S. delphini</i> (dolphins)	VP -	-	+	R	NA	NA
<i>S. hyicus</i> (pigs)	VP -	-	V	R	R	-
<i>S. lugdunensis</i>	OD +	V	-	R	R	+
<i>S. schleiferi</i>	OD -	+	-+	R	S	+
<i>S. saprophyticus</i>	urine; novobiocin (5 µg) - R; urease +; non-hemolytic	-	-	R	S	-
<i>S. epidermidis</i>	urease + non-hemolytic	-	-	R	R	-
<i>S. haemolyticus</i>	urease -, VP +, DNase -	-	-	R	S	+
<i>S. caprae</i>	urease +, DNase +	-	-	R	S	+
Other coagulase- negative staphylococci	novobiocin (5 µg) -/V; urease V; non-hemolytic or delayed hemolysis	-	-	R	S	V

+ = greater than 90% of strain positive in 48h; - = greater than 90% of strain negative; V = result are between 90 and 10% positive; -+ = most strains are negative but rare positive strains exist; NA = not applicable or available; VP = Voges-Proskauer; OD = ornithine decarboxylase; PYR = production of pyrrolidonyl arylamidase; R = resistant; S = susceptible. Bacitracin susceptible = inhibition zone 10-25 mm, Novobiocin resistant = inhibition zone ≤16 mm, Polymycin B resistant = inhibition zone <10 mm

^a Including *Kytococcus* and *Kocuria*.



ตารางที่ 5.1-2 ปฏิบัติวิธีที่ใช้ในการวินิจฉัยเชื้อ *Streptococcus* group ต่างๆ

Organisms	Group A	Group B	Non Group A, B or D	Group D <i>Enterococcus</i>	Group D not <i>Enterococcus</i>	Viridans group	<i>S. pneumoniae</i>
Hemolysis	β	β	β	α , β or γ	α	α	α
PYR (pyrrolidonyl aminopeptidase)	+	-	-	+	-	-	-
Bacitracin susceptibility 0.04 unit (at any zone size)	+	_*	_*	-	-	_*	-
CAMP reaction	-	+	-	-	-	_*	-
Bile esculin hydrolysis	-	-	-	+	+	_*	-
Tolerance to 6.5% NaCl	-	-	-	+	-	-	-
Optochin susceptibility (≥ 14 mm)	-	-	-	-	-	-	+

* บางครั้งอาจให้ผลบวก



ตารางที่ 5.1-4 การวินิจฉัยเชื้อ *Streptococcus* กลุ่ม non-pyogenic ด้วยปฏิกิริยาทางชีวเคมี

Organisms	VP	Bile esculin	Esculin	Arginine	Mannitol	Sorbitol	Urea	Optochin	Lancefield group	Remark
<i>S. suis</i>	-	v -	+	v +	-	v -	-	-	R, S, T	มีประวัติการ รับประทานหมูเค็ม หรืออาหารที่มี ส่วนผสมของหมูเค็ม หรือเลือดหมู หรือมีการสัมผัส กับสุกรหรือเนื้อหมู
<i>S. sanguinis</i> group	-	-	+	+	-	v -	-	-	None	
<i>S. mitis</i> group	-	-	- *	-	-	-	-	-	None	
<i>S. pneumoniae</i>	-	-	-	-	-	-	-	+	None	ควรยืนยันด้วย bile solubility test ถ้าพบ zone มี เส้นผ่านศูนย์กลาง น้อยกว่า 14 mm
<i>S. mutans</i> group	+	-	+	-	+	+	-	-	None	
<i>S. salivarius</i> group	+	-	v	-	-	-	v	-	None	
<i>S. anginosus</i> group	+	-	+	+	-	-	-	-	A, C, G, F, None	
<i>S. bovis</i> group	+	+++	+	-	- *	-	-	-	D	

+ = positive reaction ($\geq 95\%$); - = negative reaction ($\leq 5\%$); v + = variable reaction, but most strains

positive; v - = variable reaction, but most strains negative; v = variable reaction

* บางครั้งอาจให้ผลบวก ; ** บางสายพันธุ์อาจให้ผลลบ (*S. infantarius*)



ตารางที่ 5.1-6 การวินิจฉัยเชื้อแบคทีเรีย gram-positive cocci, catalase negative, PYR positive ที่ไม่ใช่

Streptococcus group A

Organisms	Motility	Pigment	MGP	AD	Arabinose	Mannitol	Lactose	Raffinose	Sorbitol	Tellurite	Pyruvate	45 °C Growth	Vancomycin
<i>E. avium</i>	-	-	V	-	+	+	+	-	+	-	+	+	S
<i>E. raffinosus</i>	-	-	V	-	+	+	+	+	+	-	+	+	V
<i>Vagococcus fluvialis</i>	+	-	+	-	-	+	-	-	+	-	-	-	S
<i>E. faecium</i>	-	-	-	+	+	+d	+	V	V	-	-	+	V
<i>E. gallinarum</i>	+d	-	+	+d	+	+	+	+	D	-	-	+	R
<i>E. casseliflavus</i>	+d	+d	+	+d	+	+	+	D	V	-d	V	+	R
<i>E. mundtii</i>	-	+	-	+	+	+	+	+	V	-	-	+	S
<i>E. faecalis</i>	-	-	-	+d	-	+d	+d	-	+	+	+	+	V
<i>Lactococcus garvicæ</i>	-	-	-	+	-	+	+	-	-	-	-	-	S
<i>E. durans</i>	-	-	-	+	-	-	+	-	-	-	-	+	S
<i>E. hirae</i>	-	-	-	+	-	-	+	+	-	-	-	+	S
<i>E. dispar</i>	-	-	+	+	-	-	+	+	-	-	+	-	S



A scenic autumn path lined with trees with golden foliage. The path is a light-colored gravel or dirt road, covered with fallen orange and yellow leaves. It leads into the distance, flanked by green grass and rows of trees whose leaves are in various stages of autumn color, from bright yellow to deep orange. The trees form a canopy over the path, with sunlight filtering through the leaves. The overall atmosphere is peaceful and beautiful.

Thank you for your
attention
&
Good Luck

